

王維碩

國立臺北大學 電機工程學系研究所

混和訊號積體電路設計 # 資訊彙整



關於我...

我是王維碩，來自高雄小港，畢業於國立臺北大學電機系研究所，論文內容主要為通訊IC中的鎖相迴路(PLL)設計到下線、並完成繪製PCB打線圖等流程，並於TSRI訊號量測。個性喜歡探索事物新的可能性，樂於分享自己所學，興趣是繪畫、攝影，休閒時刻喜歡騎腳踏車、慢跑，喜歡關注有關自然的議題。樂意為工作和新機會花更多時間進修，或是透過在新工作任務的作業中，提供自己的建議或看法。

學習歷程

1. 國立臺北大學 - 電機工程學系研究所 (畢業總平均: 89.43　GPA: 3.86 / 4)

2. 國立金門大學 - 電子工程學系 (畢業總平均: 87.26　系排名: 2/64　百分比: 3.13)

3. 陽明交大新尖兵 - 科技與AI應用人才培訓班 (特優成績結訓)
   課程內容: AI與數據分析、電子電路設計、半導體製程技術

4. 台灣半導體研究中心課程 (Design of RF CMOS IC、CMOS射頻積體電路(PA)設計與實作)



證照

1. Parallax Certified Professional
2. TQC 人工智慧應用與技術
3. TOEIC 多益測驗 570分
4. 救護技術員 EMT-1









技能與著作

1. 論文 : 四相位壓控振盪器設計與具延遲電路調整之5GHz鎖相迴路
2. 專題 : 以深度學習為基礎之人工智慧探討
3. PCB 電路板設計與元件焊接
4. 簡報設計與報告撰寫

經歷

國立金門大學 – 教學助理 (2019年)

科目: 微積分、通訊原理(協助監考與試卷批改)

國立金門大學 – 計畫助理 (2020年)

計畫課程需進行計畫講義排版美化、簡報資料彙整、實驗室設備管理、電腦軟體維護與裝修、上課拍照及錄影與後製剪接。

國立臺北大學 – 計畫助理 (2021 – 2022年)

計畫資料彙整、計畫經費管理與申報、公文跑流程簽核、子電路施作、計畫成果報告書資料圖形製作、實驗室設備財產管理、電腦軟體維護與裝修(含公文操作系統流程說明製作、繪圖與模擬軟體操作流程製作)。

經歷



財團法人嘉義市私立瑞泰老人長期照護中心 - 替代役職員 (2023年5月至10月)

內勤任務:
維持戶外庭院環境整潔(掃地、澆花)、派車單資料處理(輸入隔日需看診住民之資料、需求、與費用計算)、氧氣瓶罐注與確認。

外勤任務(醫院約4家):
整院檢體遞送、至醫院取回整院藥品、伴護住民至醫院就醫(含陪同身心障礙鑑定、臨時急診)、接出院(推擔架床至醫院伴護接回住民)、機構喘息個案伴護。

論文研究

四相位壓控振盪器設計與具延遲電路調整之 5GHz 鎖相迴路
Quadrature-VCO Design and 5GHz Phase-Locked Loop with Controlled Delay Line



行動通訊技術中的收發機架構中，為了使其中之振盪器具備更大頻寬及更佳的抗雜訊能力,故使用鎖相迴路系統的技術來實行,系統中使用的電路有相位頻率偵測器、充電幫浦、低通濾波器、壓控振盪器和除頻器,為了配合大規模巨量天線(MIMO)技術可能應用,故對於晶片內部需要的控制因素增加,於此論文中嘗試在除頻器輸出至相位頻率偵測器之間加入具延遲功能之可變電容電路(Varactor Delay Element)來增加系統可調性,且於 PCB 板上放置兩塊晶片嘗試組合成多輸出多控制的調整目的。

研究生 : 王維碩
指導教授 : 林嘉泫 博士

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- Introduction
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- Phase Frequency Detector & Charge pump
- Loop Filter
- Voltage Controlled Oscillator
- Quadrature-Voltage Controlled Oscillator
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- Layout & Measurement
- Conclusion & Reference

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5G New Radio in Release 16[15][16]

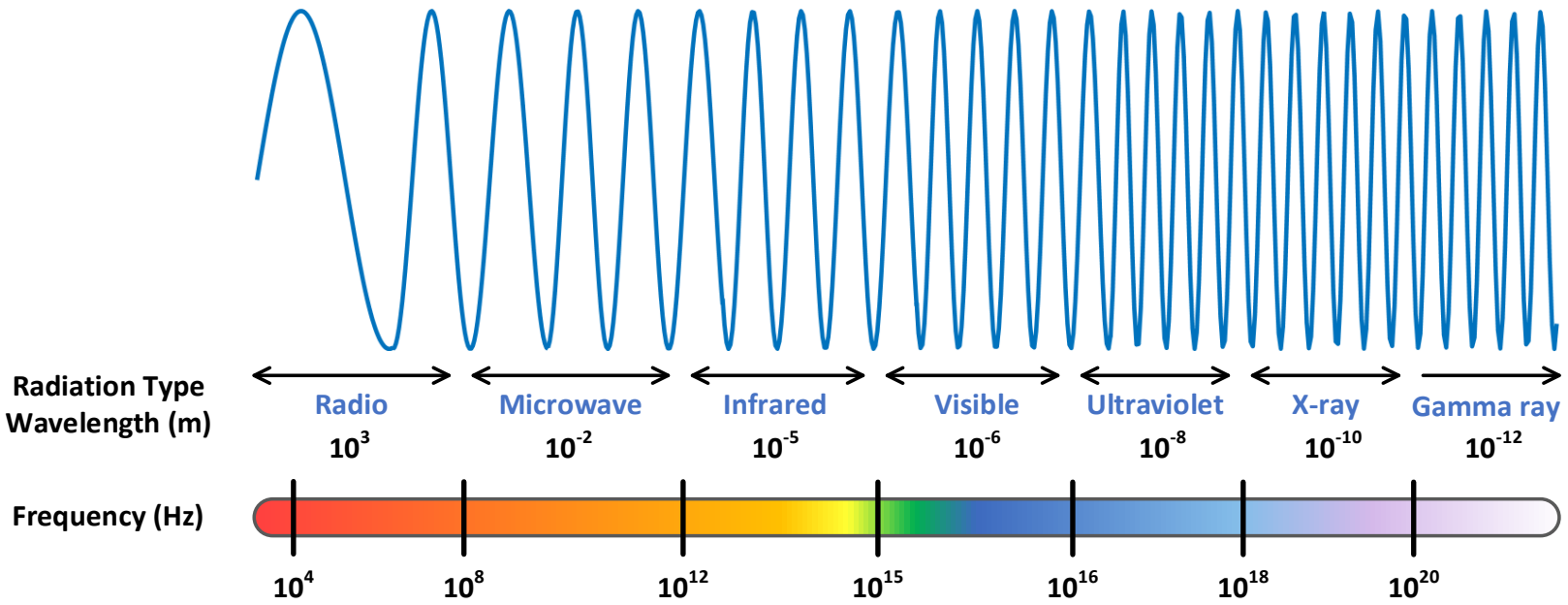


Fig 1. Electromagnetic waves

Tab 1. 3GPP 5G NR FR1 add to Release 16

Frequency band	*f*（MHz）	Upload（MHz）	Download（MHz）
n46	5200	5150 – 5925	
n96	6000	5925 – 7125	

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Superheterodyne Receiver Structure

Local Oscillator

- Generate a stable signal for mixers for mixing frequency.
- If the signal is not stable, it will be easily changed by noise and the frequency shake is too large.
- Providing a stable LO signal requires the use of a phase-locked loop system architecture.

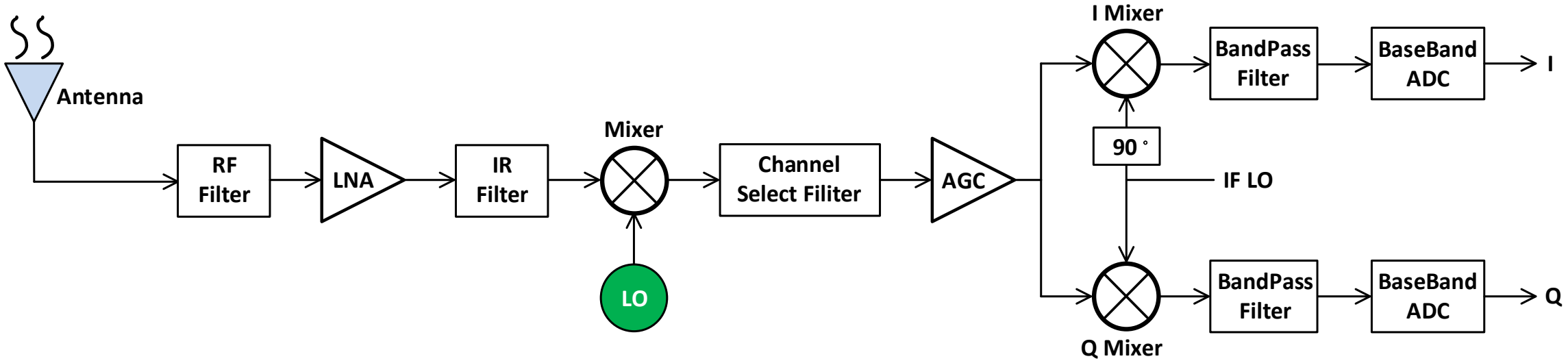


Fig 2. Structure of superheterodyne receiver

Phase-Locked Loop

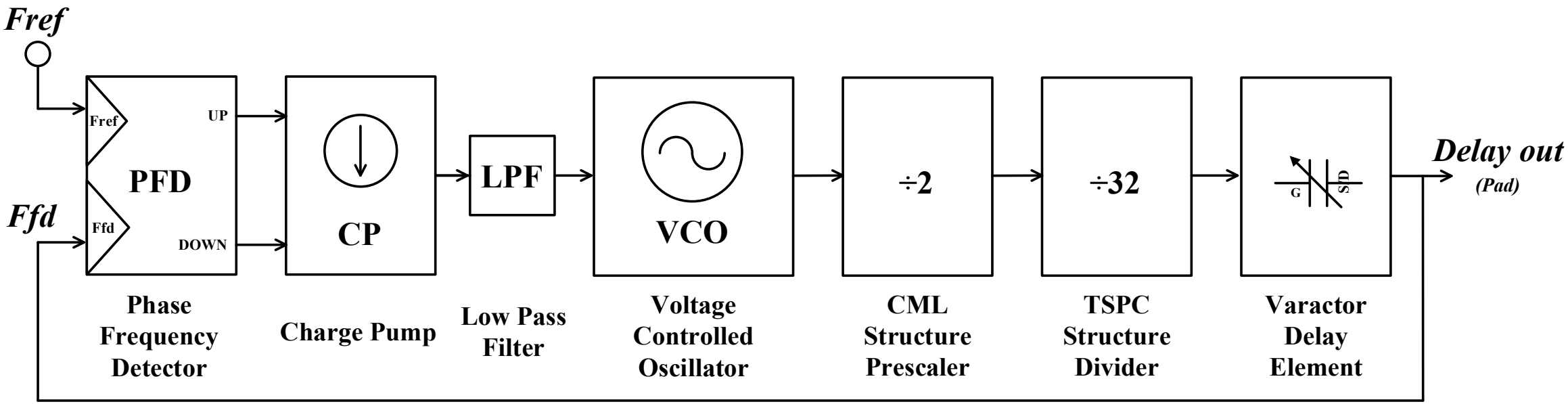


Fig 3. System of phase-locked loop

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Phase Frequency Detector

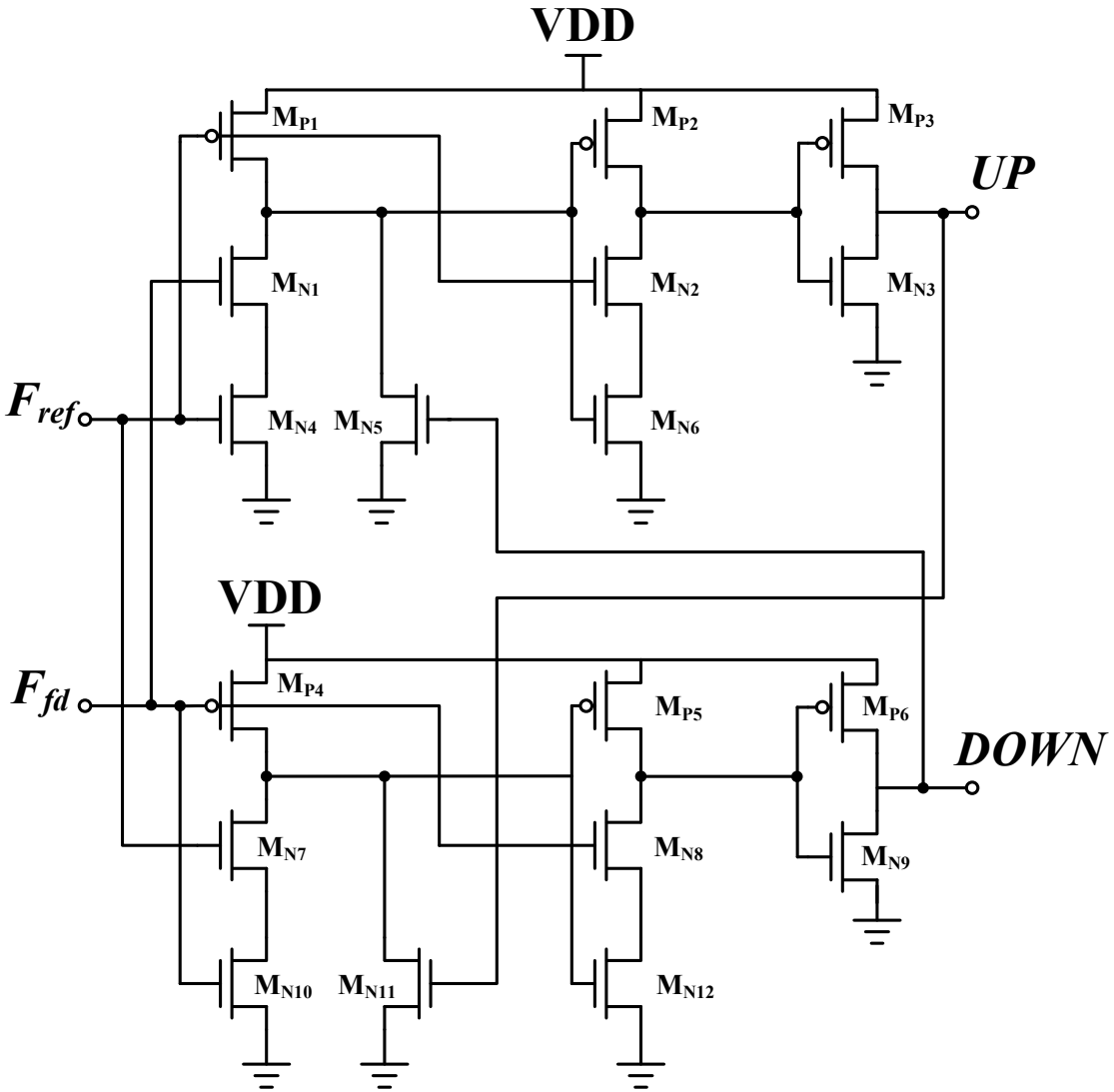


Fig 4. Structure of phase frequency detector

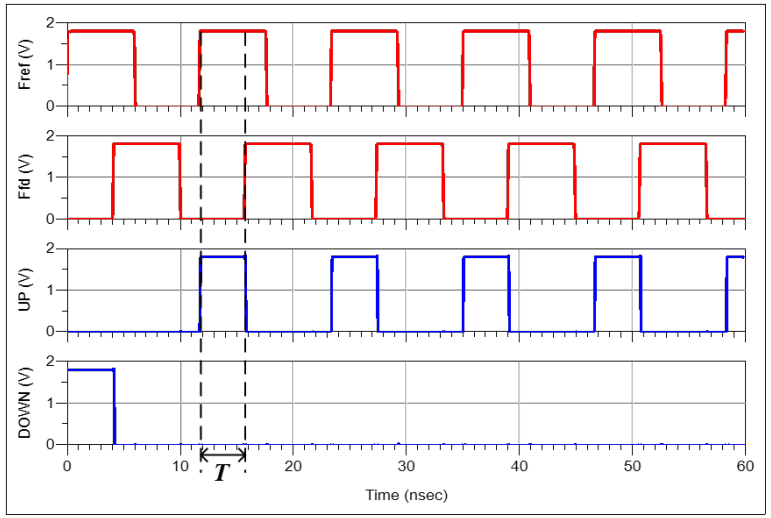


Fig 5. F_{ref} lead to F_{fd}

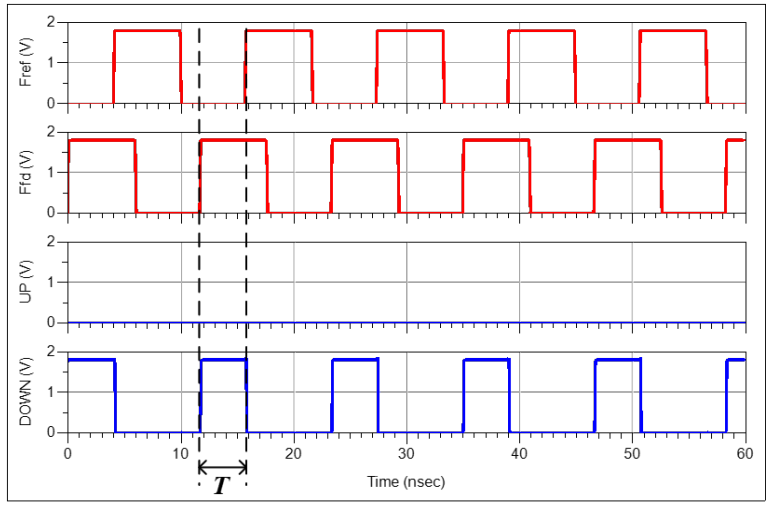


Fig 6. F_{fd} lead to F_{ref}

Phase Frequency Detector

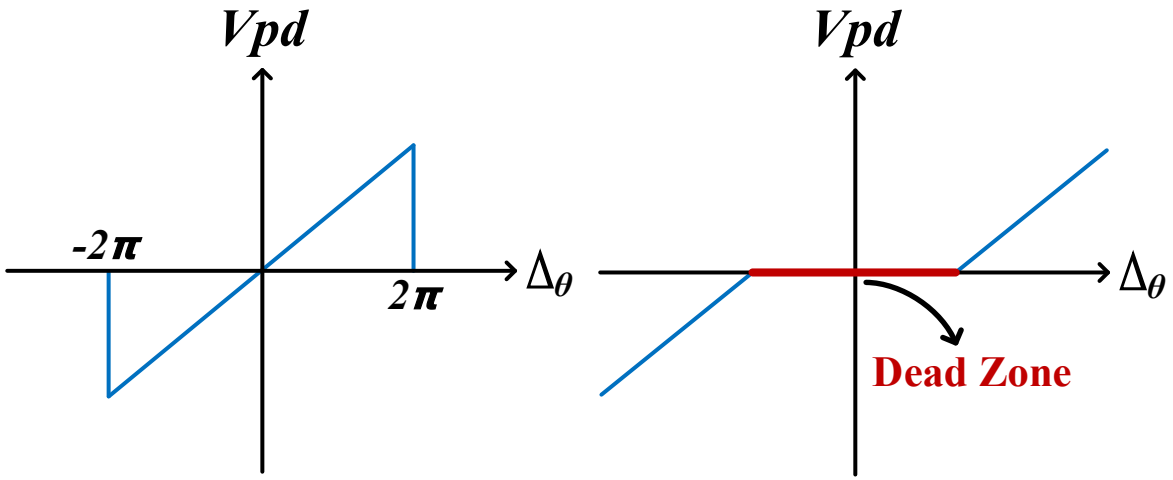


Fig 7. Dead Zone

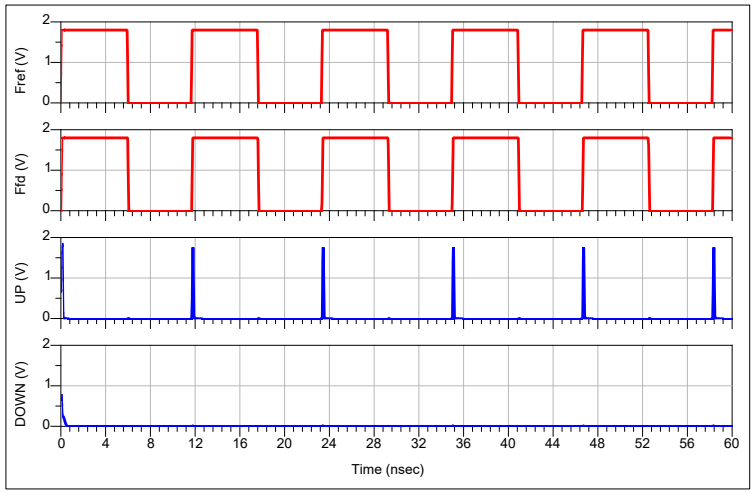


Fig 8. F_{ref} lead 0.05 nsec to F_{fd}

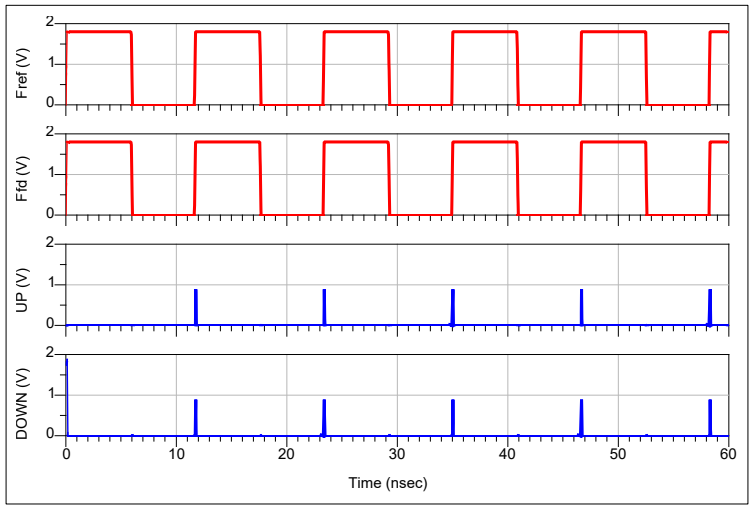


Fig 9. F_{ref} equal to F_{fd}

Charge Pump

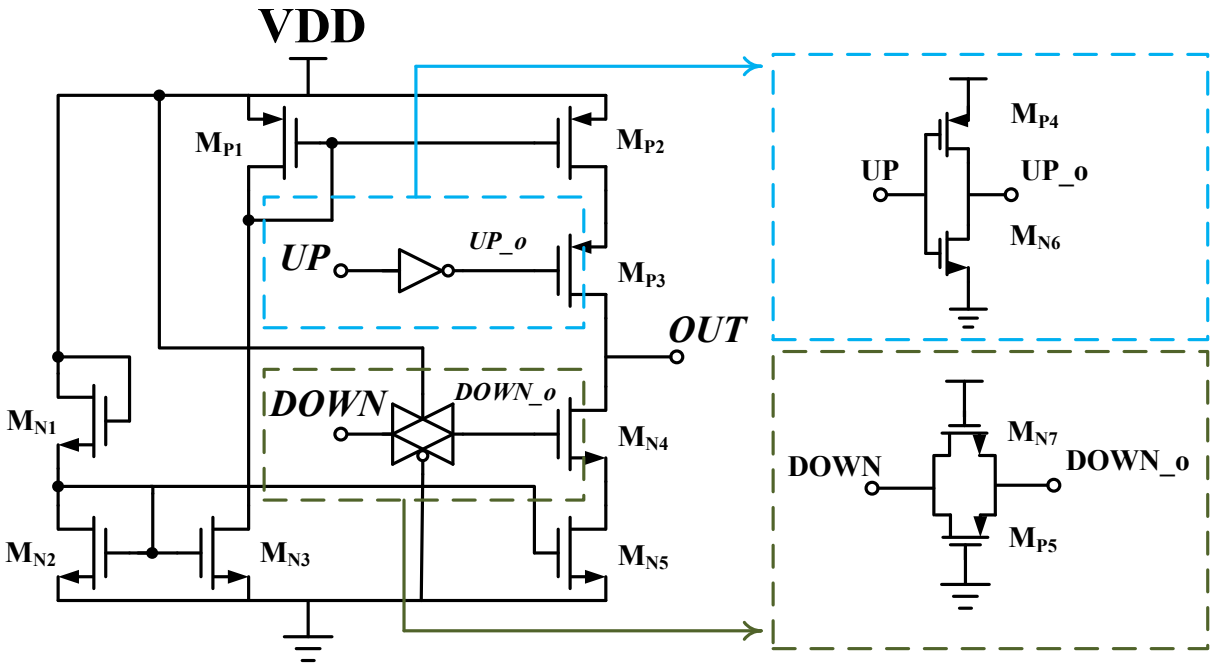


Fig 10. Structure of Charge Pump

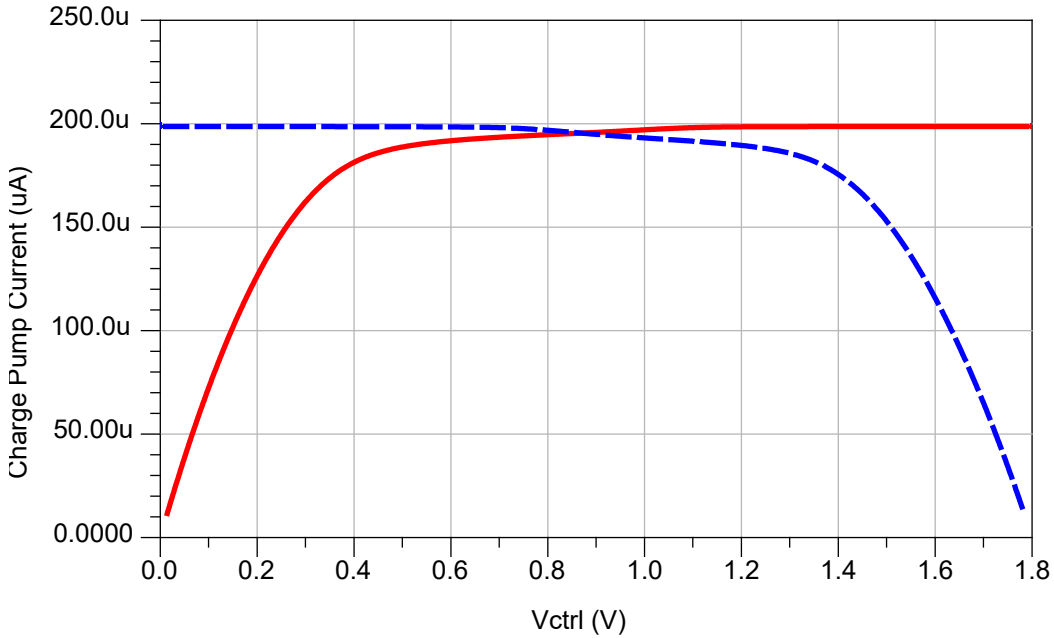


Fig 11. Charge pump charge/discharge current curve

Charge pump

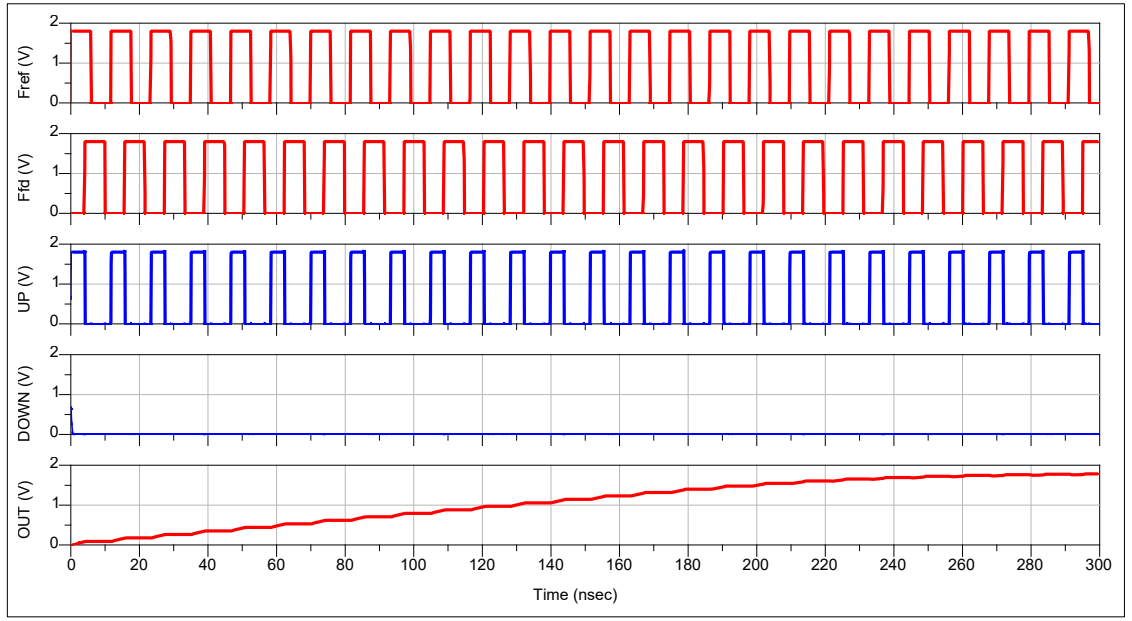


Fig 12. Charge

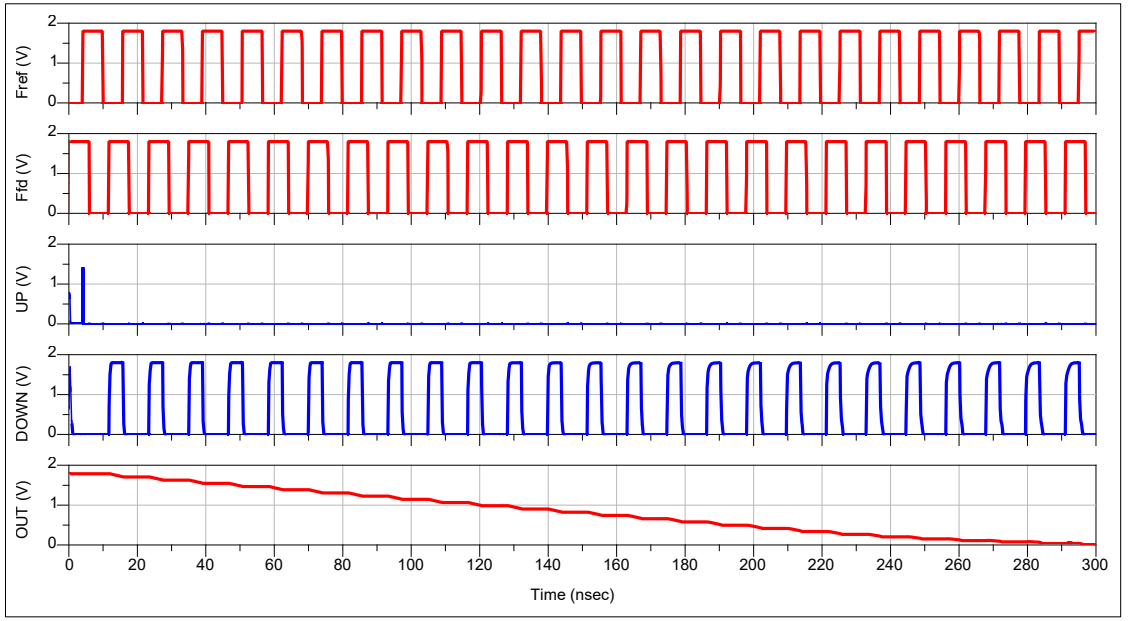


Fig 13. Discharge

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Structure of low-pass filter

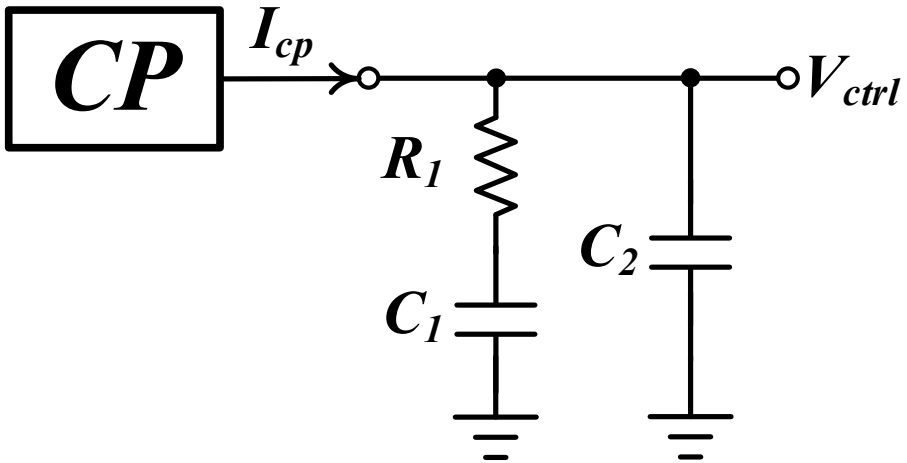


Fig 14. Structure of low-pass filter

$$Z(s) = K_{lpf} \cdot \frac{s + \omega_2}{s \cdot \left(1 + \frac{s}{\omega_3}\right)}$$

$$K_{lpf} = \frac{R_1 C_1}{C_1 + C_2}$$

$$\omega_3 = \frac{C_1 + C_2}{R_1 C_1 C_2} = \omega_2 \cdot \left(1 + \frac{C_1}{C_2}\right)$$

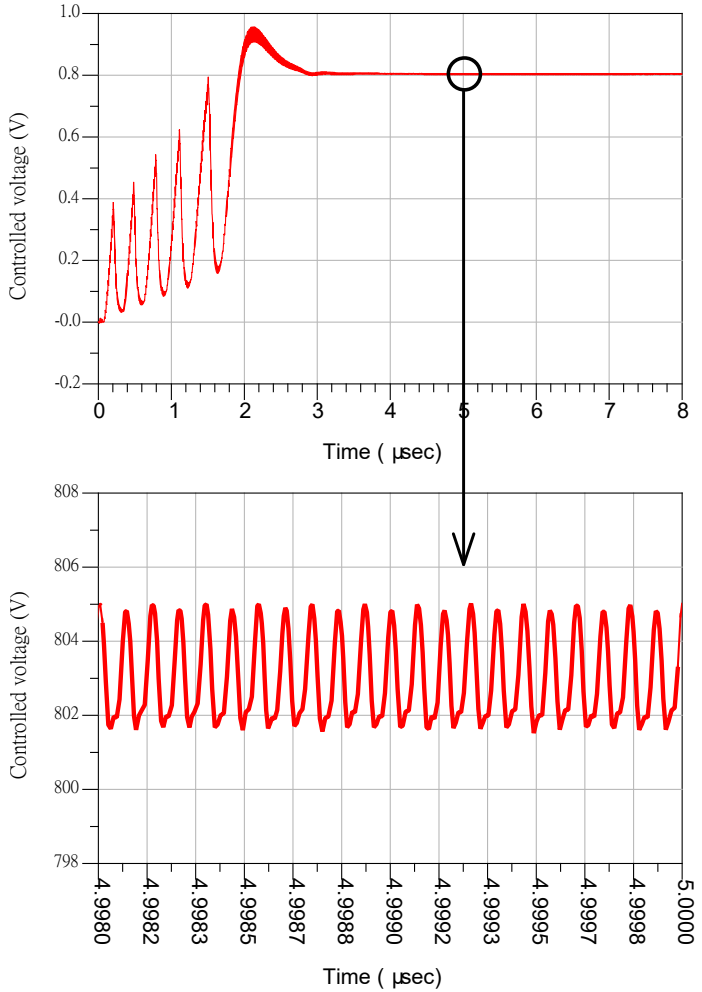


Fig 15. Function of low-pass filter

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Structure of Voltage Controlled Oscillator

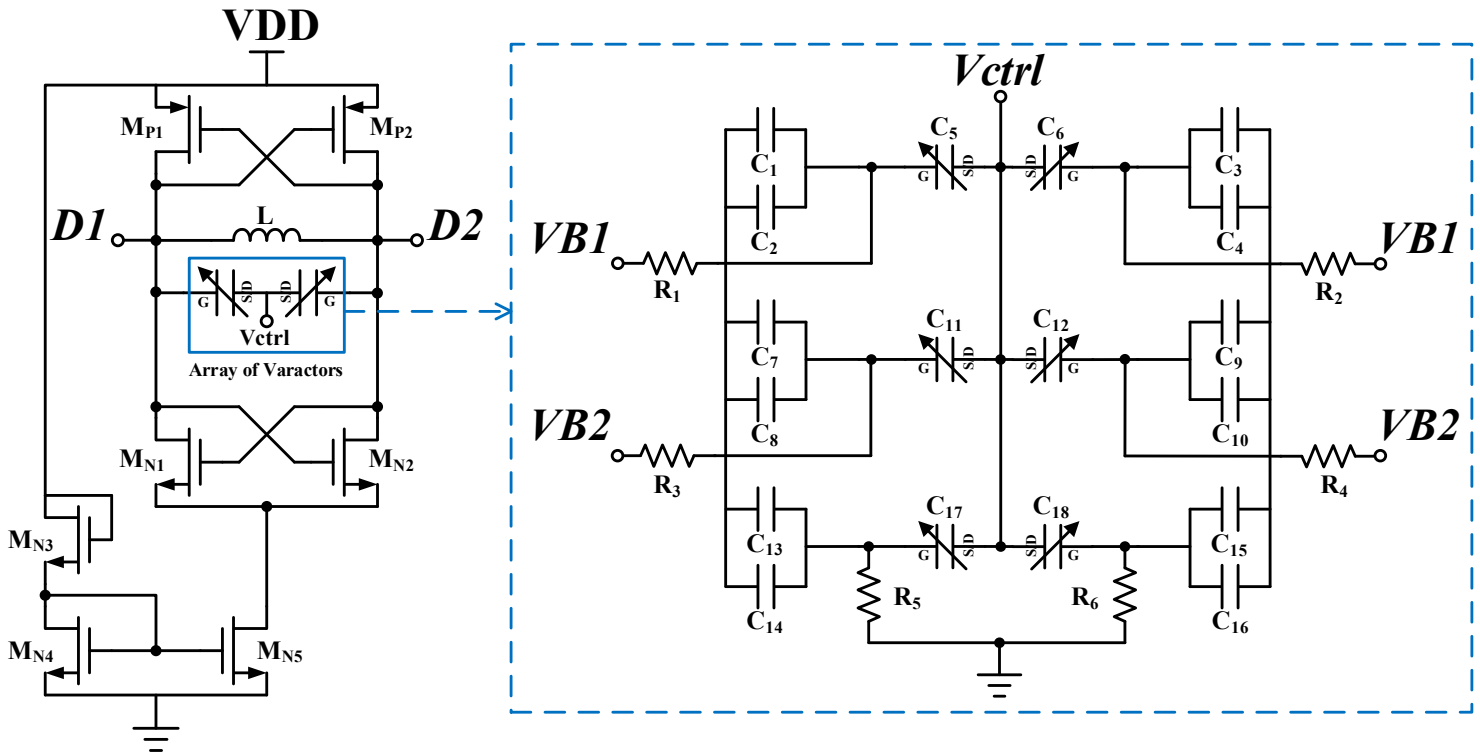


Fig 16. Structure of voltage-controlled oscillator

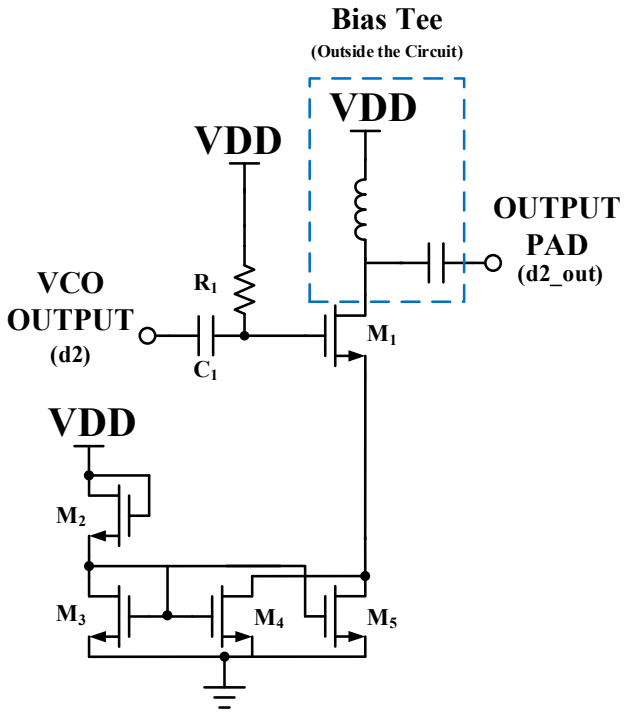


Fig 17. VCO buffer circuit

Voltage Controlled Oscillator Output Signal

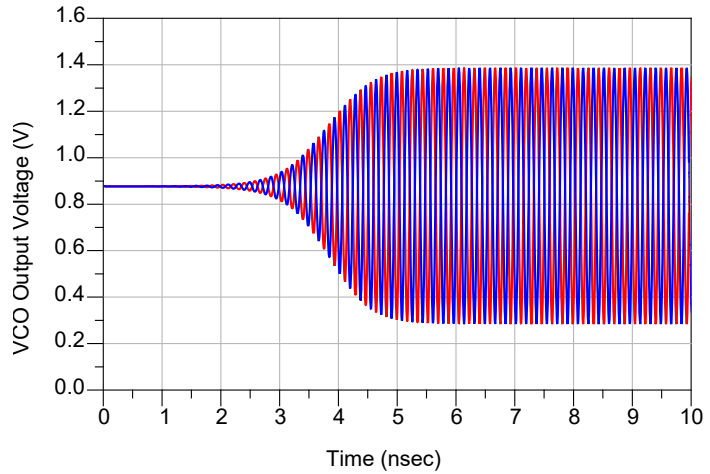


Fig 18. VCO circuit start-up time

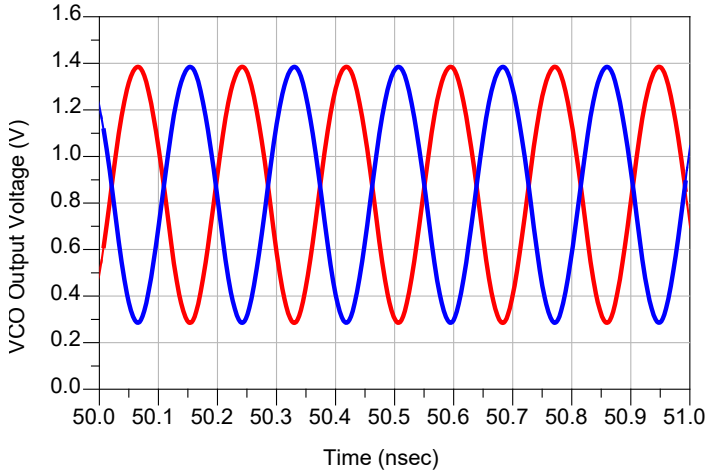


Fig 19. VCO circuit output

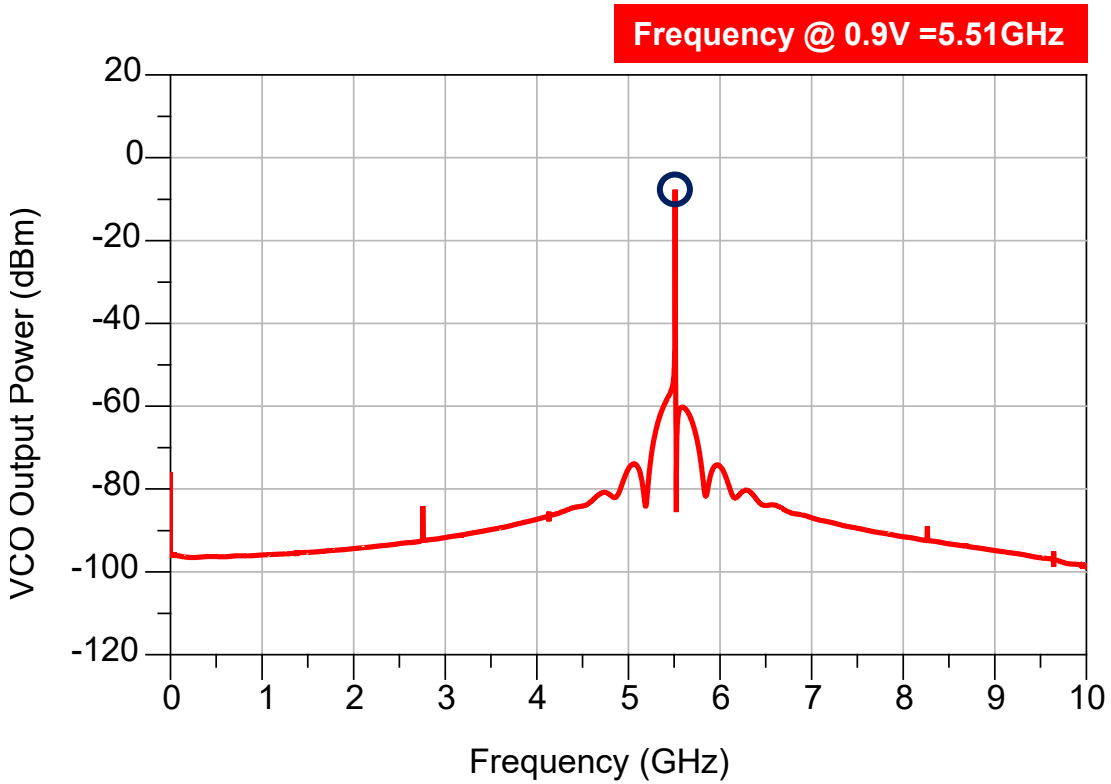


Fig 20. VCO circuit output after Fourier transform

Voltage Controlled Oscillator Output Signal

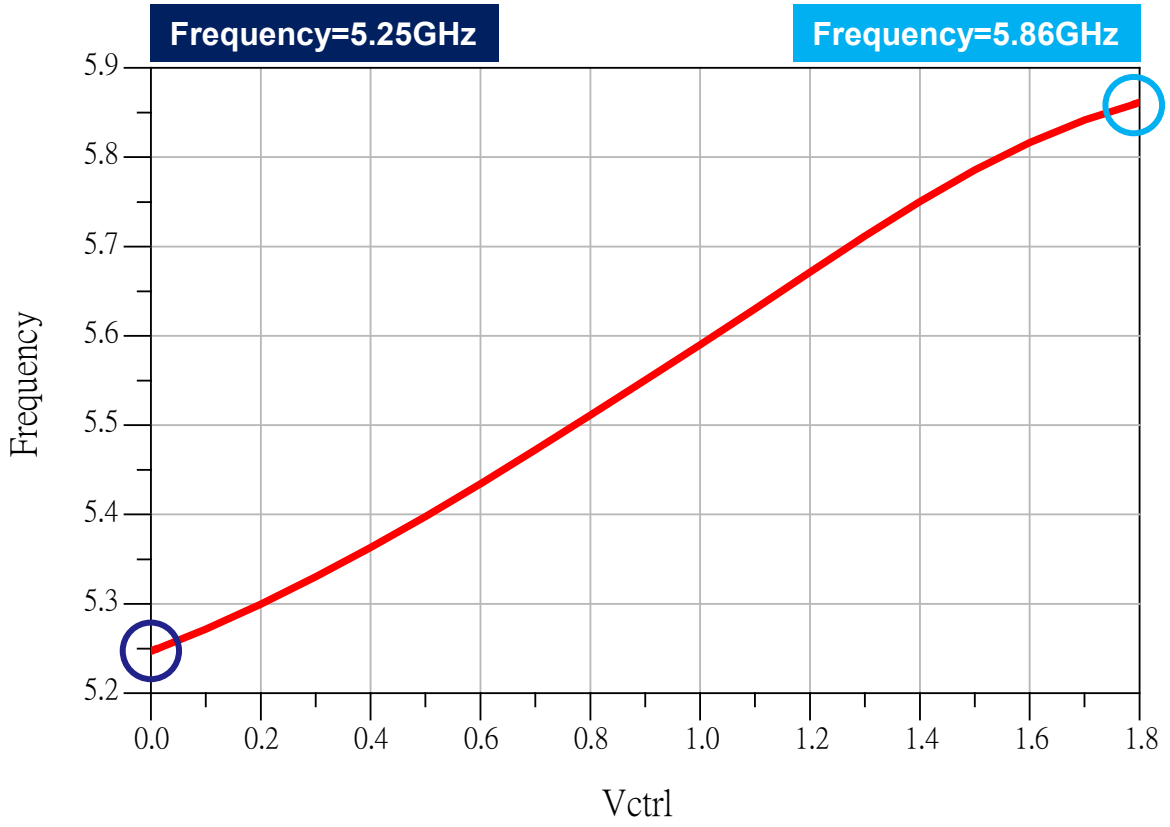


Fig 21. Tuning range of VCO circuit

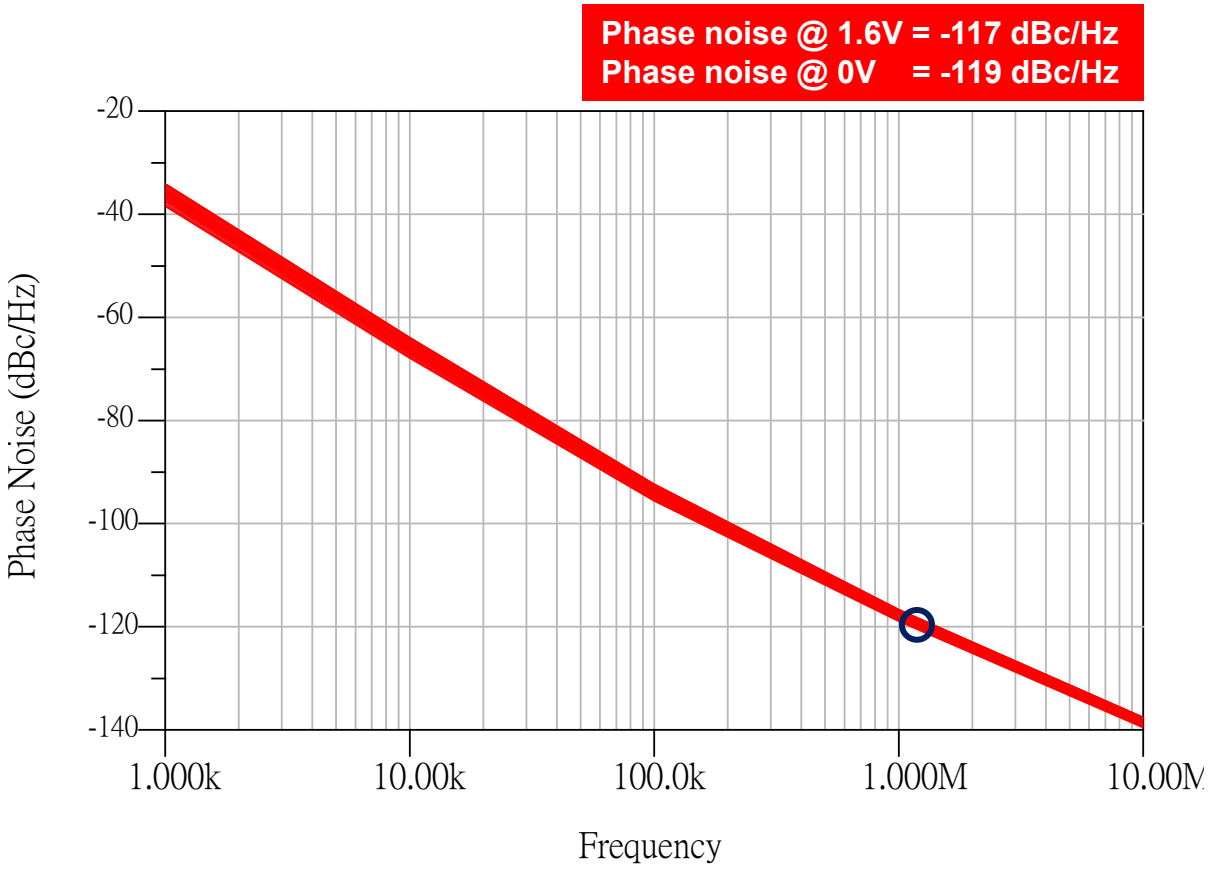


Fig 22. Phase noise of VCO circuit

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Structure of Quadrature Voltage Controlled Oscillator

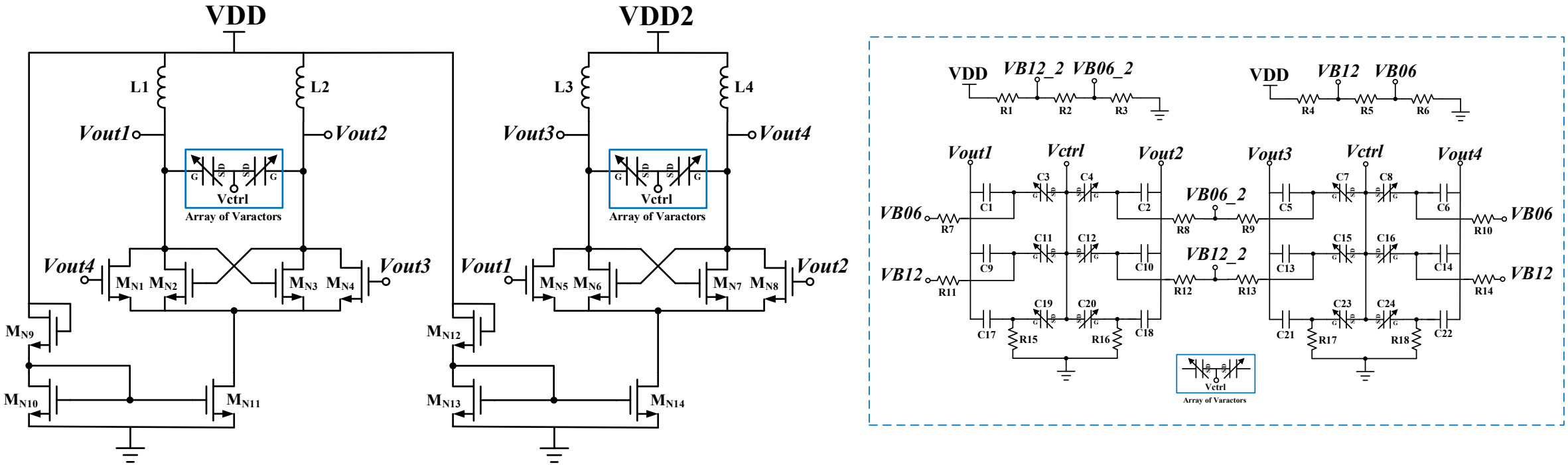


Fig 23. Structure of quadrature voltage controlled oscillator

Simulation of QVCO

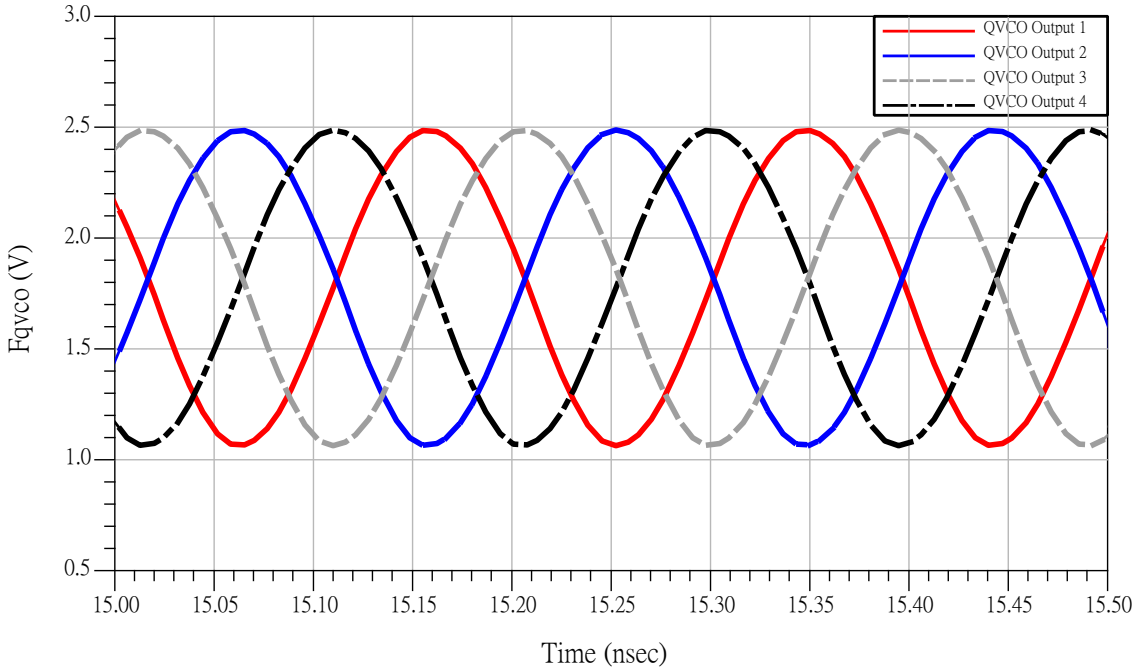


Fig 24. QVCO circuit output

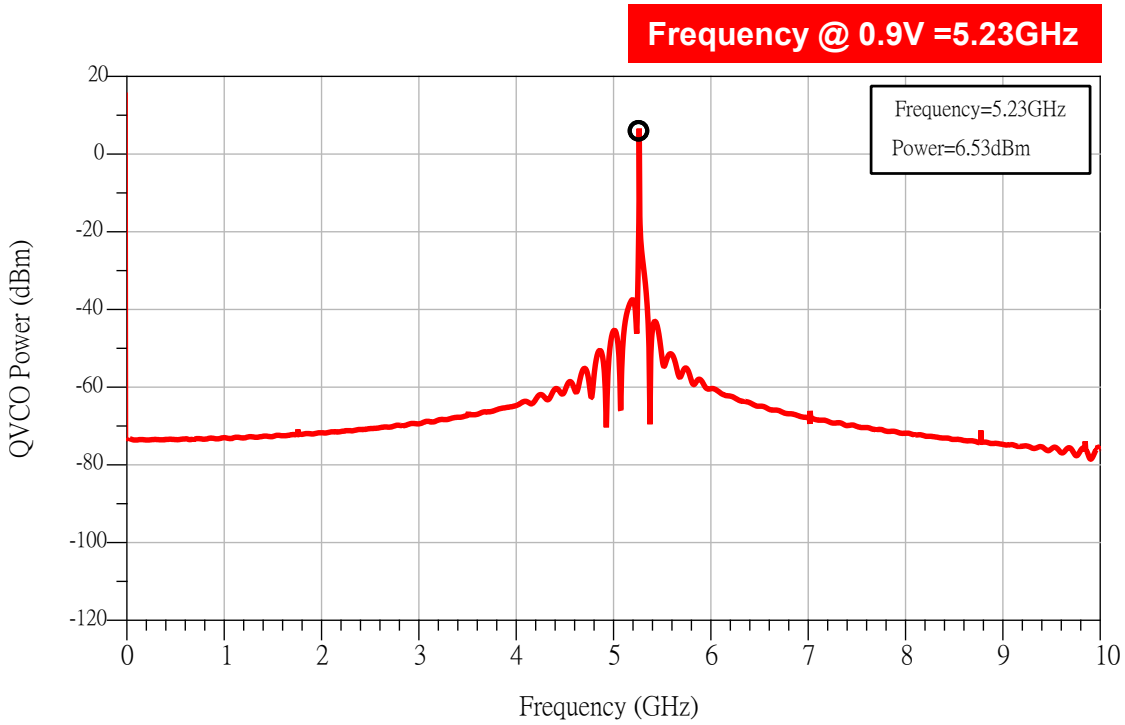


Fig 25. QVCO circuit output after Fourier transform

Simulation of QVCO

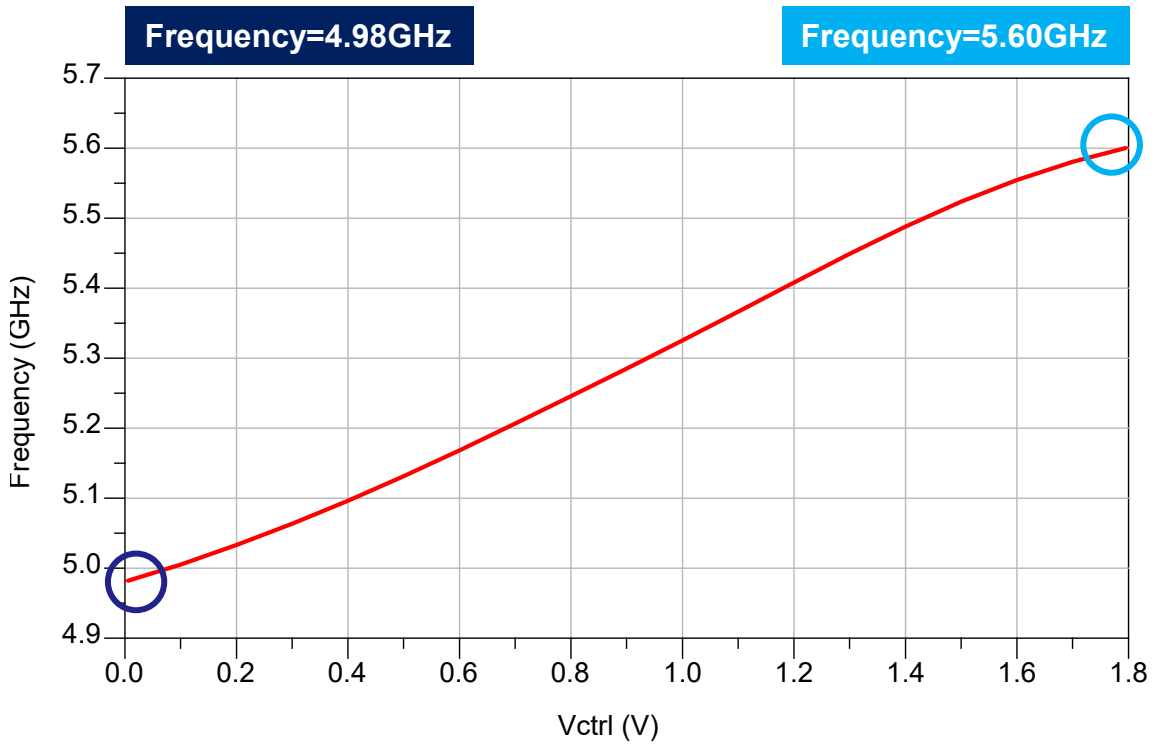


Fig 26. Tuning range of VCO circuit

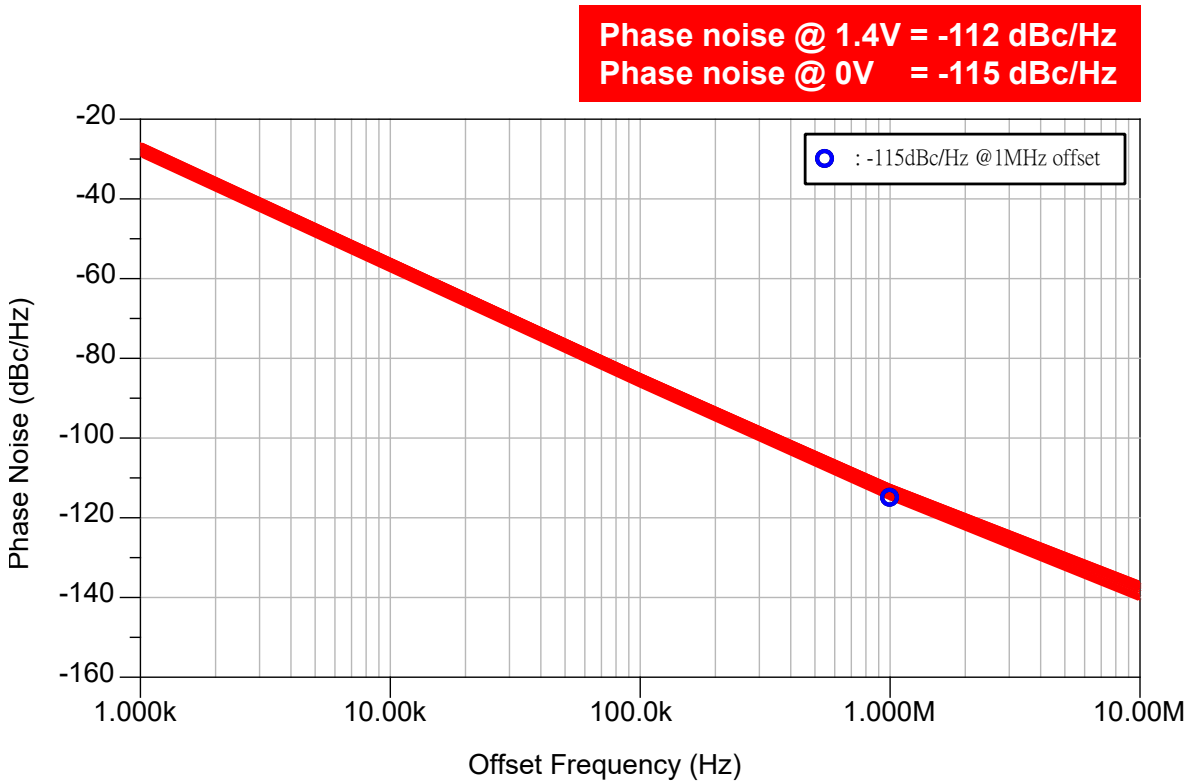


Fig 27. Phase noise of QVCO circuit

Phase Noise

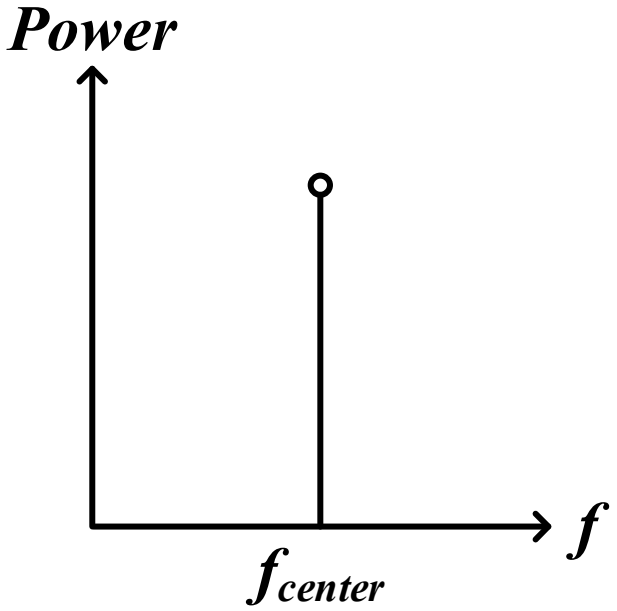


Fig 28. VCO circuit idea output signal

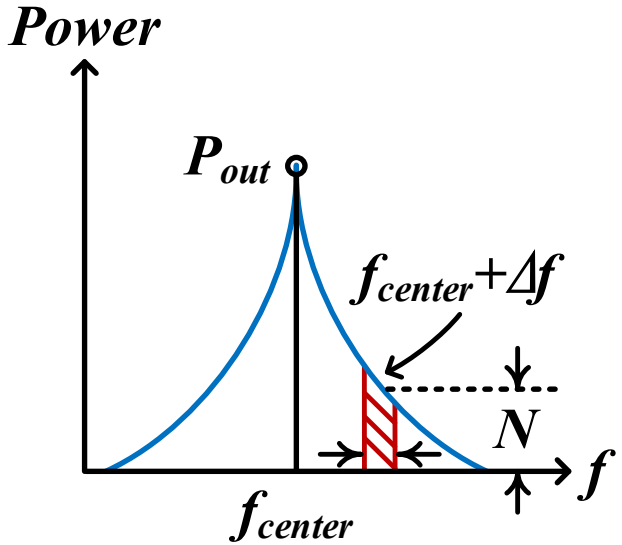


Fig 29. Real output signal

$$L\{\Delta f\} = \frac{f_{center} + \Delta f \text{ power at } 1MHz \text{ bandwidth}}{\text{carrier power}}$$

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Structure of Prescaler

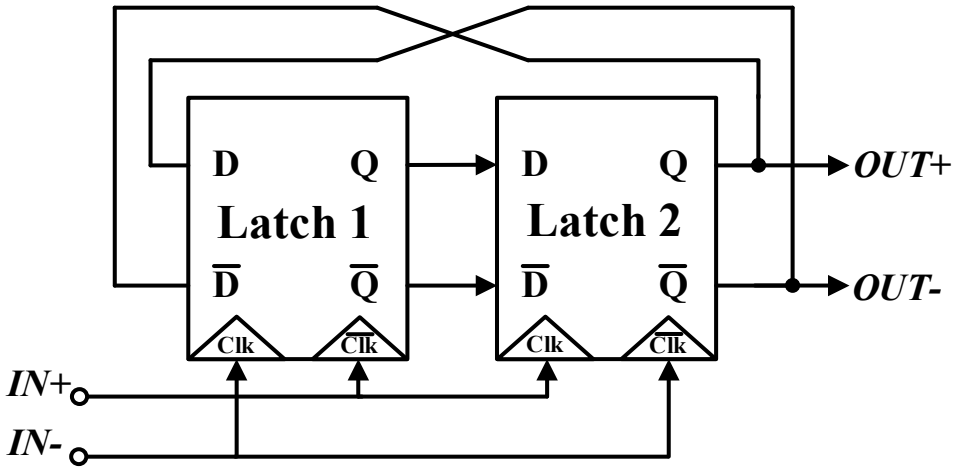


Fig 30. Structure of prescaler

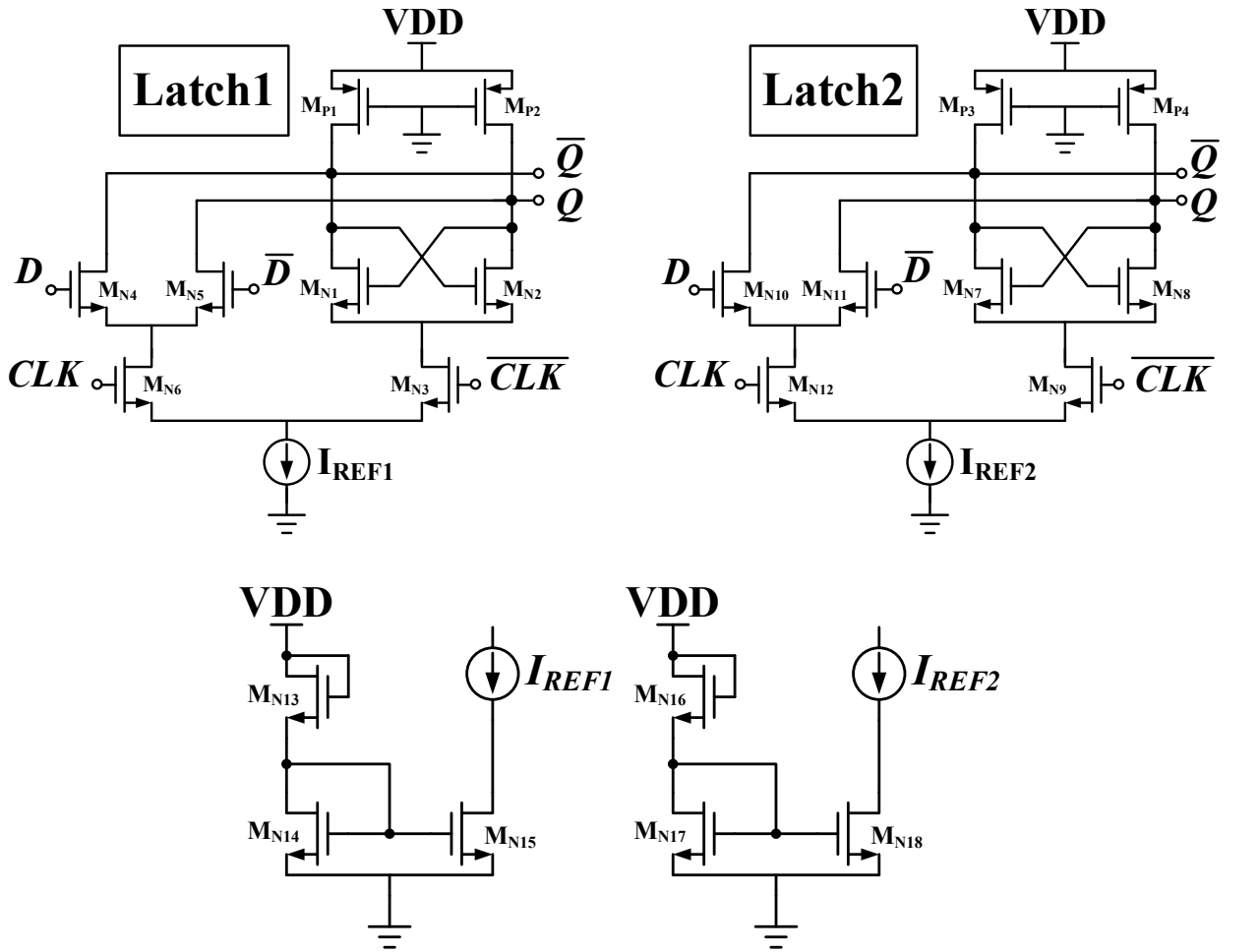


Fig 31. Detail structure of prescaler

Prescaler Simulation

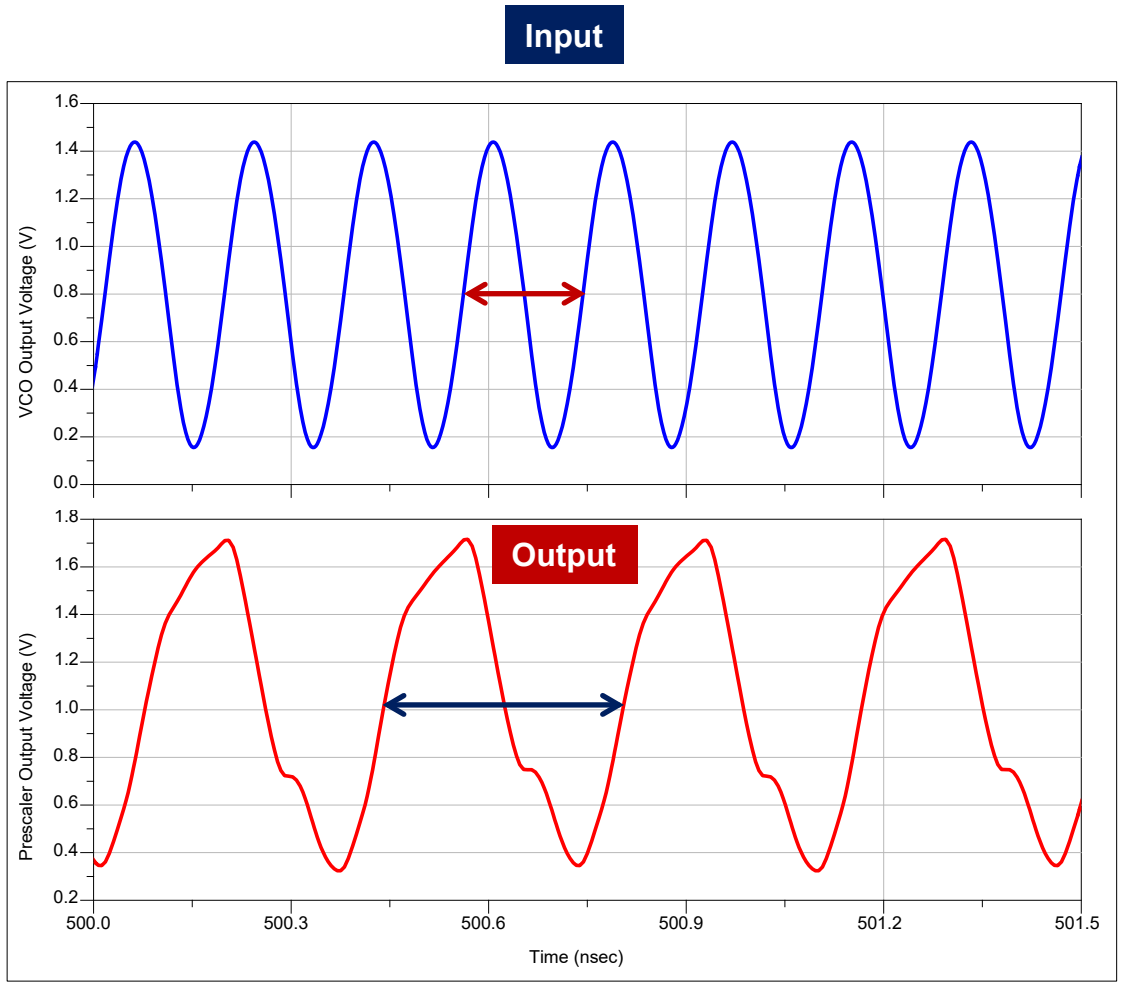


Fig 32. Input and output of prescaler

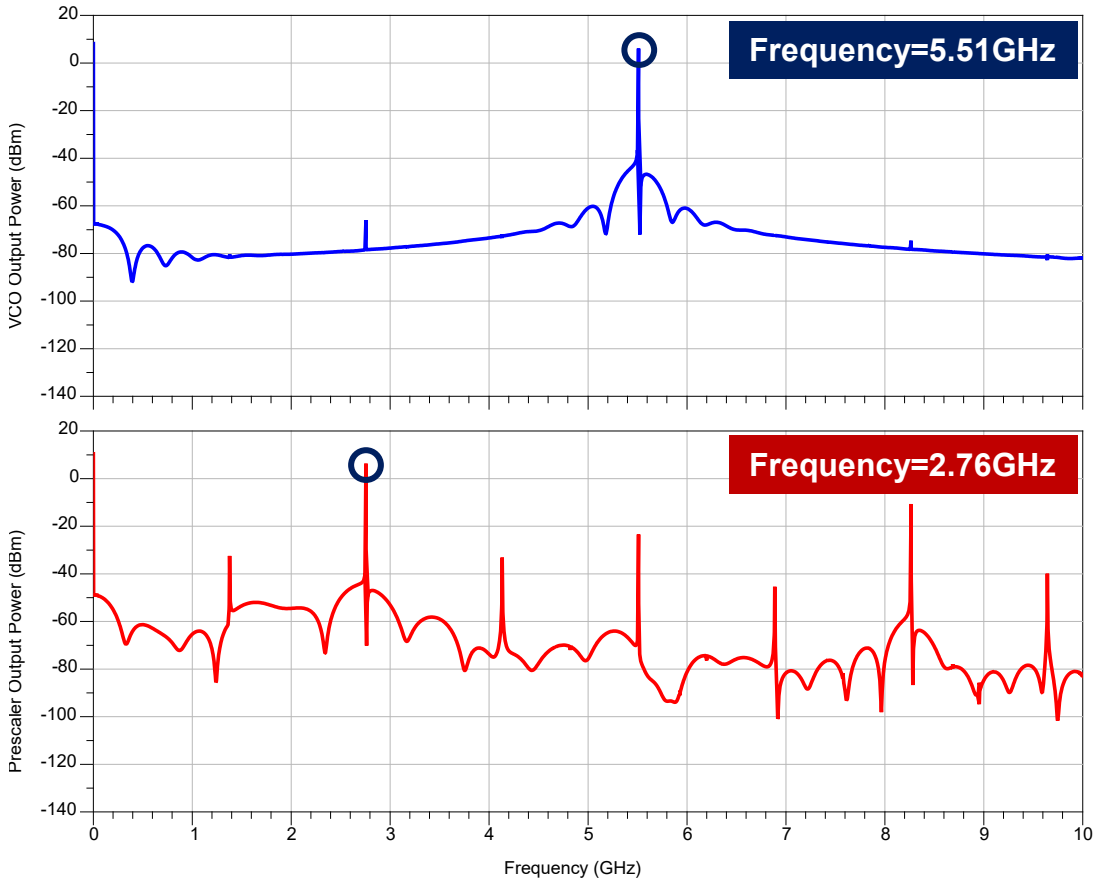


Fig 33. Input and output of prescaler with Fourier transform

Structure of True Single Phase Clock Divider

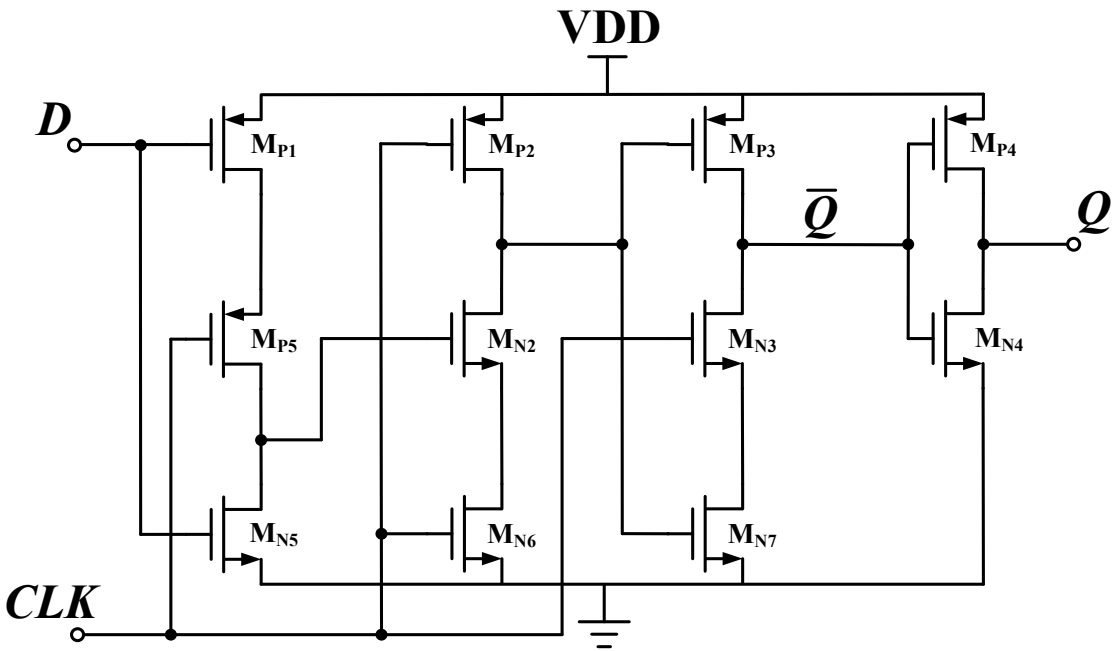


Fig 34. TSPC divider circuit

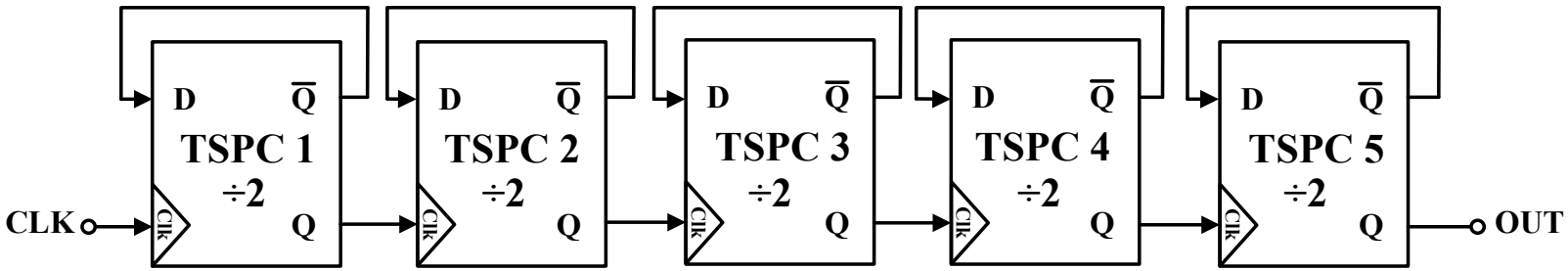


Fig 35. Chain structure of TSPC divider

TSPC Divider Chain Output

÷2
1.38 GHz

÷4
689 MHz

÷8
344 MHz

÷16
172 MHz

÷32
86 MHz

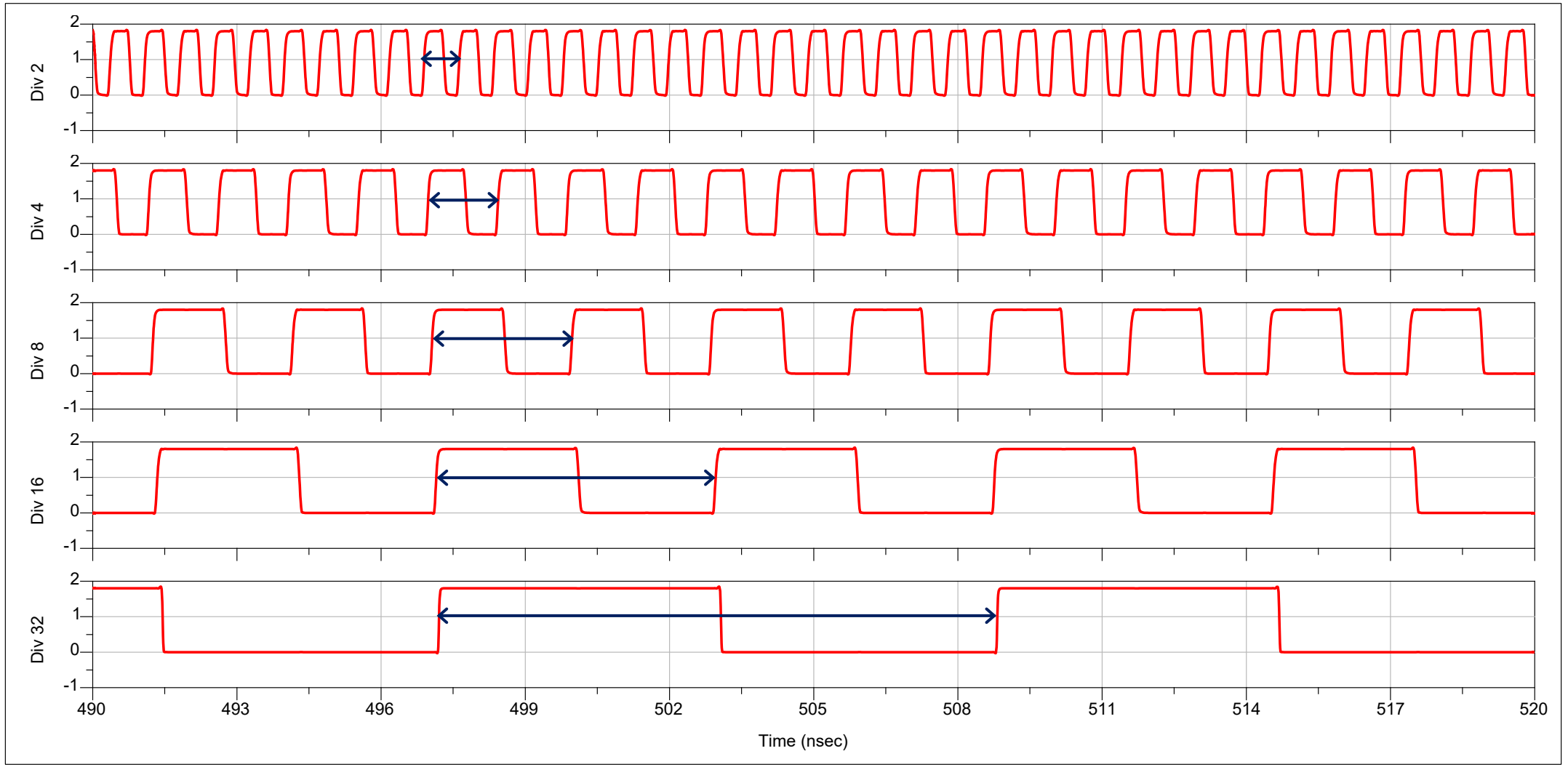


Fig 36. TSPC divider chain output signal

Contents

- Introduction
- Related Technology
- Phase Frequency Detector & Charge pump
- Loop Filter
- Voltage Controlled Oscillator
- Quadrature-Voltage Controlled Oscillator
- Frequency Divider
- **Varactor Delay Element**
- Layout & Measurement
- Conclusion & Reference

Structure of Varactor Delay Element

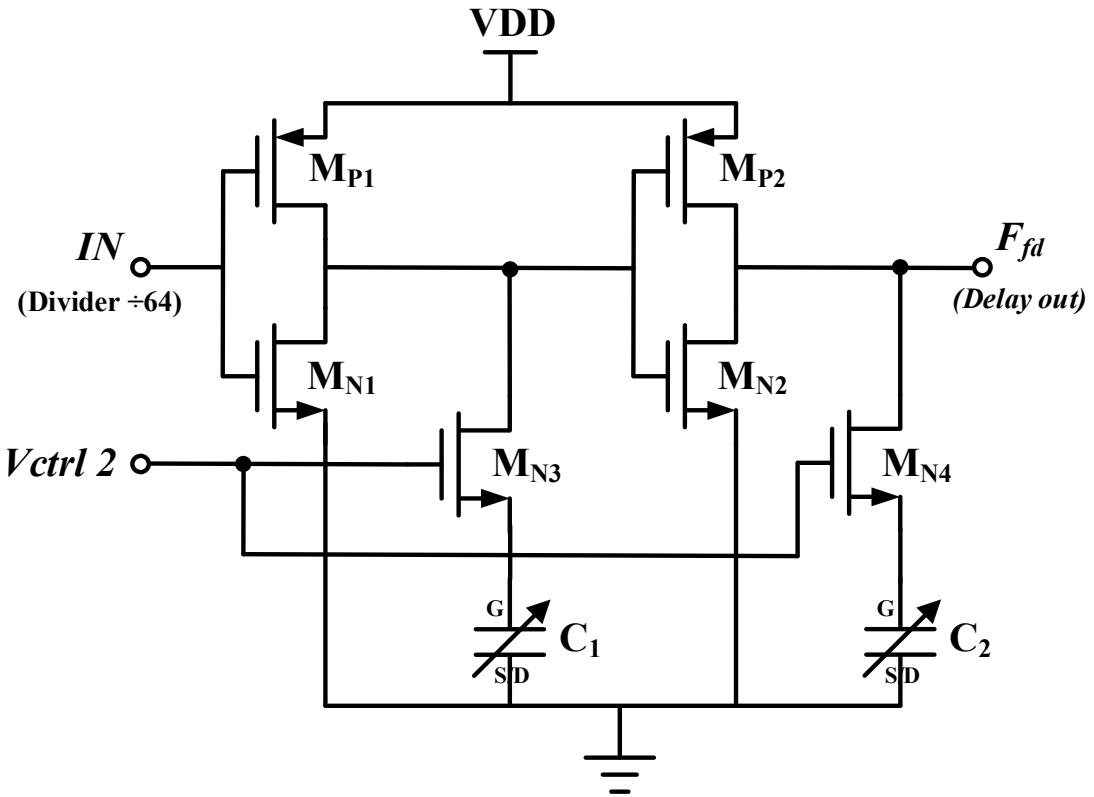


Fig 37. Structure of varactor delay element

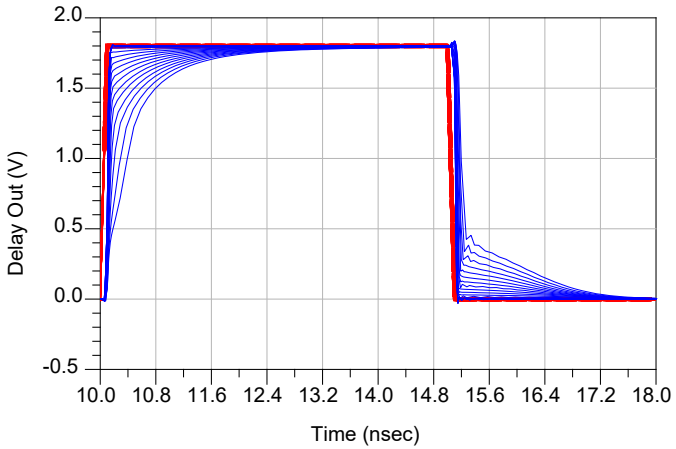


Fig 38. Delay curve of varactor

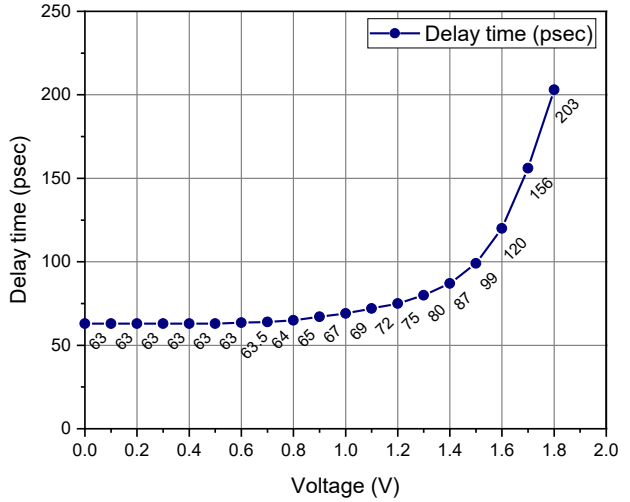


Fig 39. Delay time with control voltage

Contents

- Introduction
- Related Technology
- Phase Frequency Detector & Charge pump
- Loop Filter
- Voltage Controlled Oscillator
- Quadrature-Voltage Controlled Oscillator
- Frequency Divider
- Varactor Delay Element
- **Layout & Measurement**
- Conclusion & Reference

I/O Pin of Phase-Locked Loop System

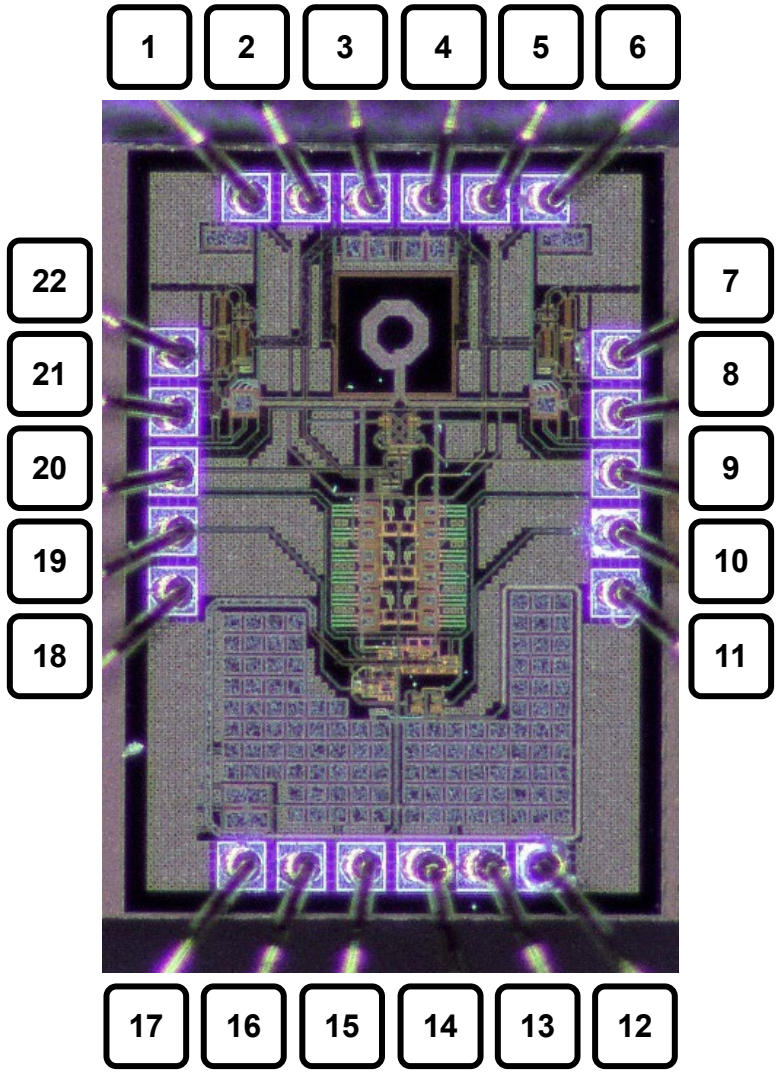
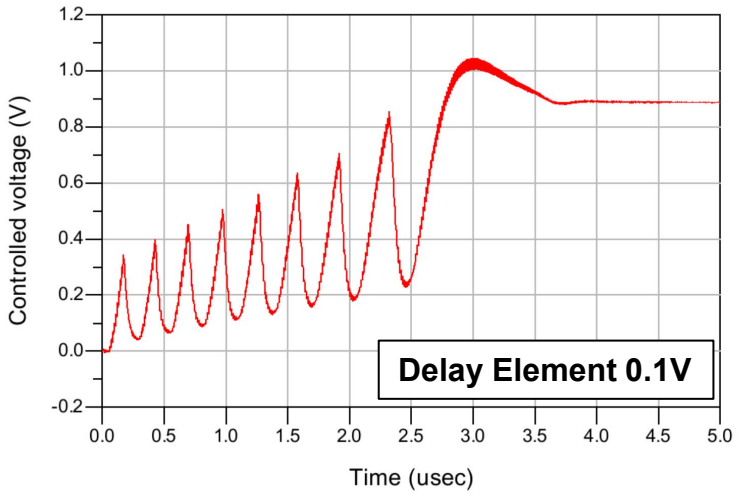


Fig 40. I/O Pin

Tab 2. I/O Pin of Phase-Locked Loop System

1	VB1	Supply for Array of Varactors	12	VDD	Supply for Digital Circuit
2	GND	Ground for chip	13	GND	Ground for chip
3	VDD	Supply for Analog Circuit	14	Vctrl	Test for Vctrl
4	VDD	Supply for Analog Circuit	15	Vctrl2	Supply for Delay Element
5	GND	Ground for chip	16	GND	Ground for chip
6	VB2	Supply for Array of Varactors	17	VDD2	Supply for Digital Circuit
7	GND	Ground for chip	18	GND	Ground for chip
8	D2	Output for VCO	19	Fref	Reference frequency
9	GND	Ground for chip	20	GND	Ground for chip
10	Ffd	Test for feed back	21	D1	Output for VCO
11	GND	Ground for chip	22	GND	Ground for chip

Post Simulation of Phase-Locked Loop System



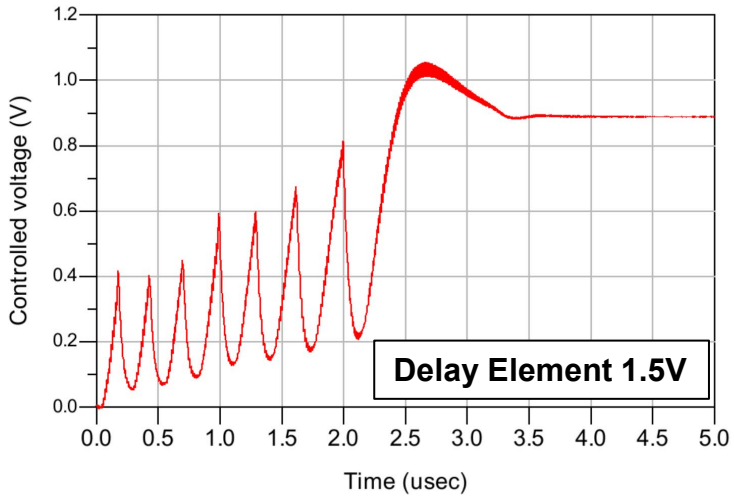
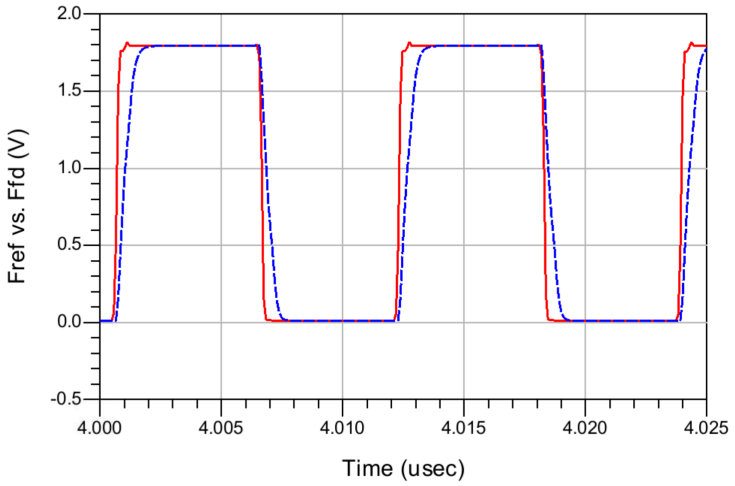


Fig 41. Phase-locked loop system control voltage



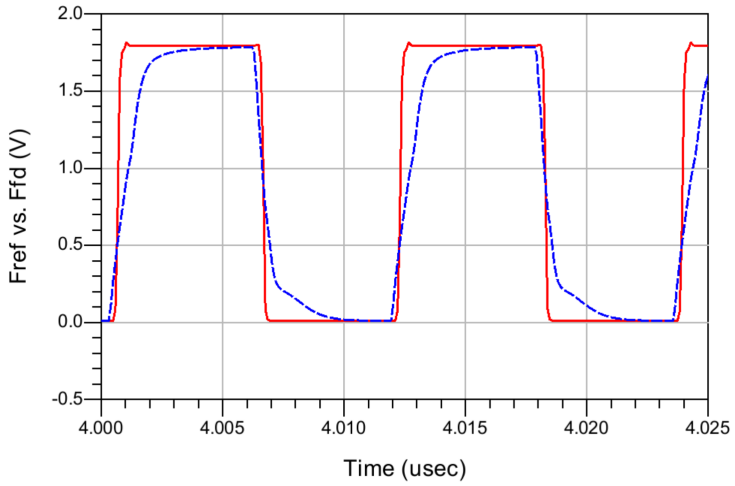


Fig 42. Feedback signal *Ffd*

Post Simulation of Phase-Locked Loop System

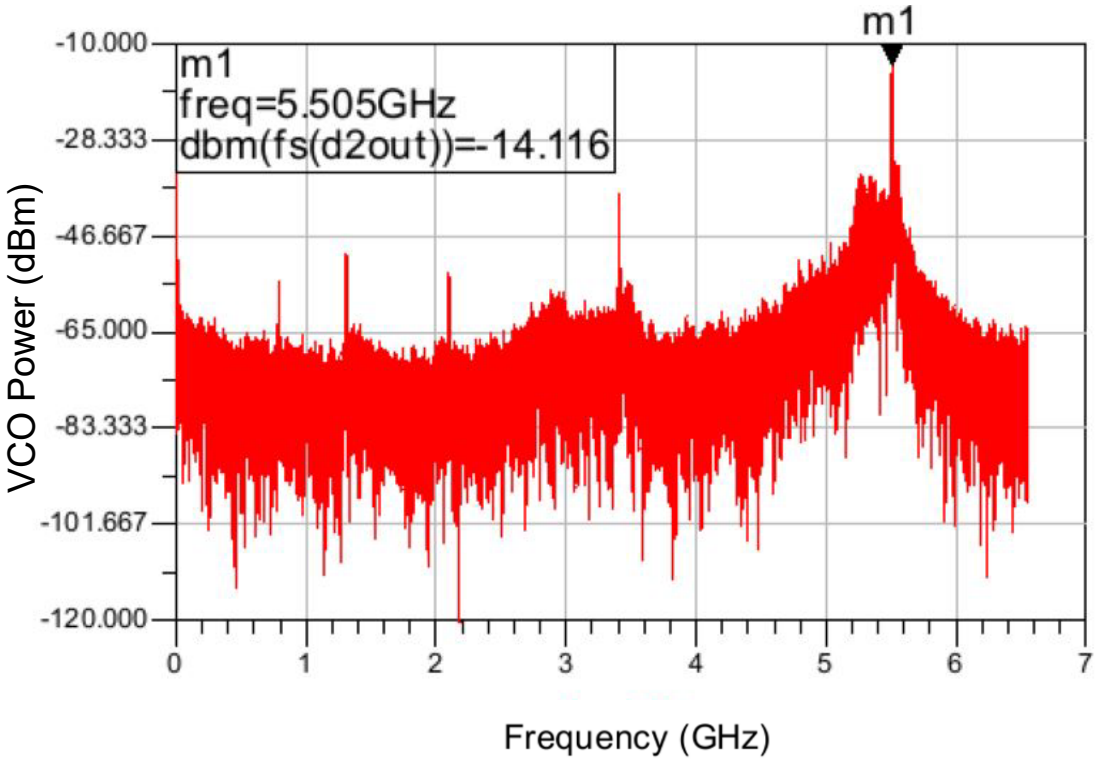


Fig 43. VCO circuit output power when PLL locked

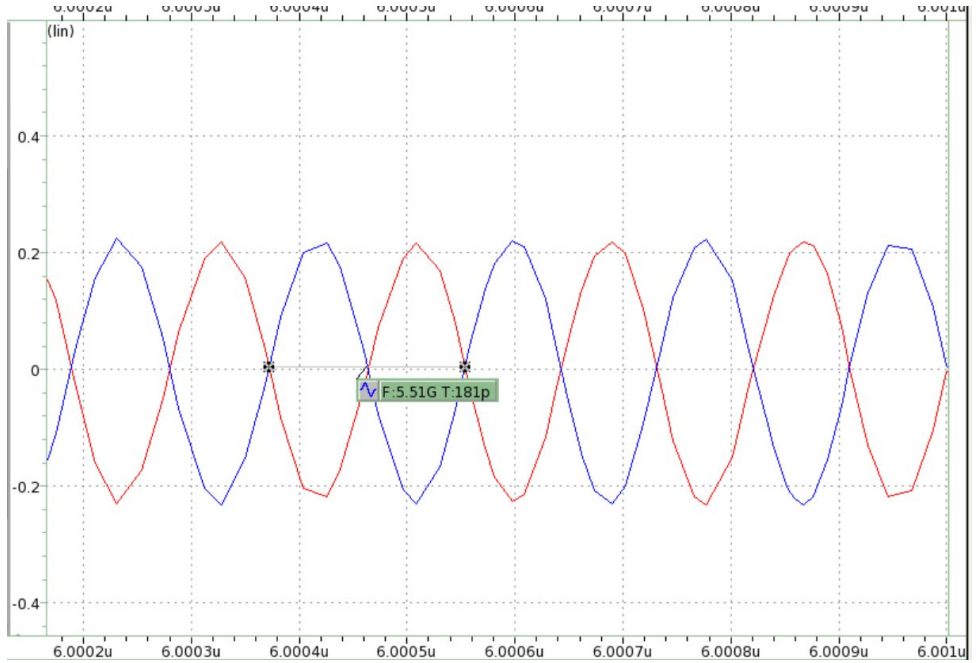


Fig 44. VCO circuit output signal when PLL locked

PCB Board with Two Phase-Locked Loop Chips

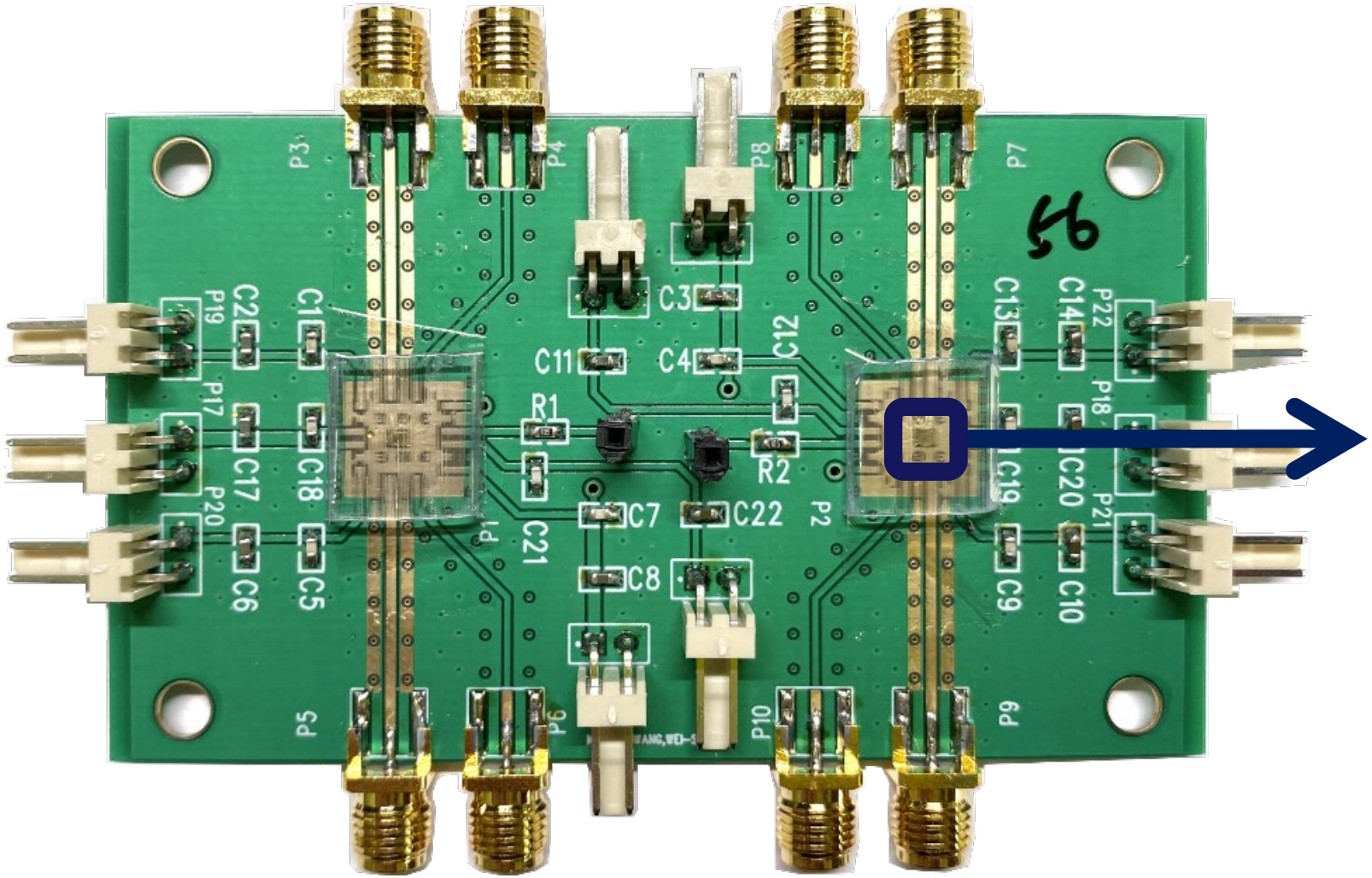


Fig 45. PCB board

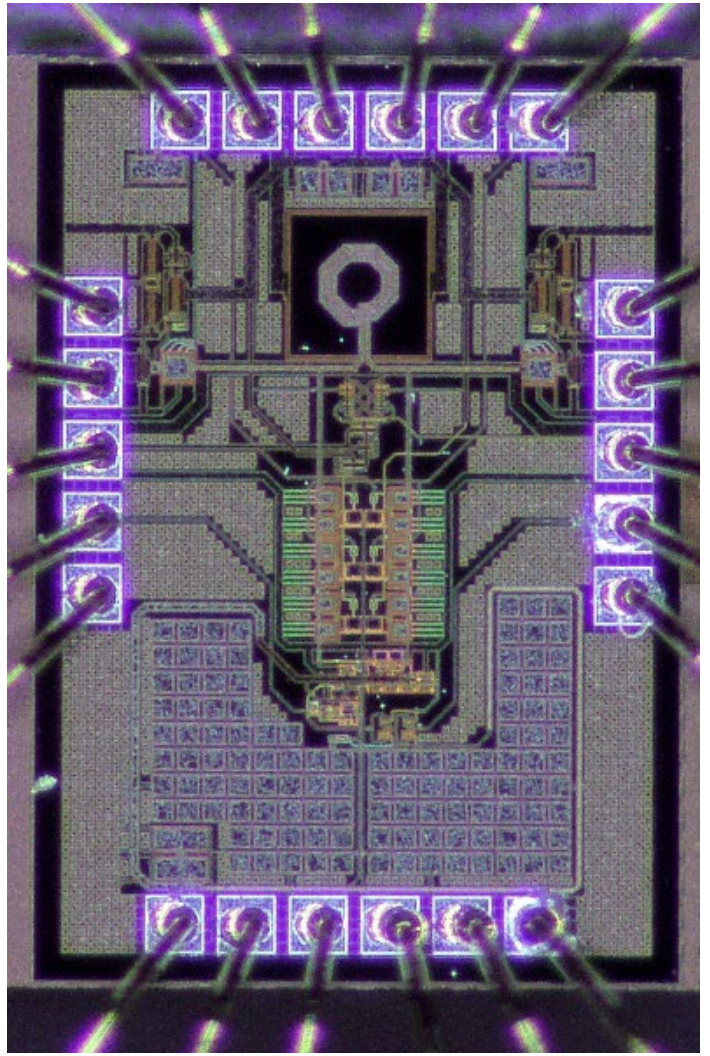


Fig 46. Photomicrograph of phase locked loop chip

PCB Board with Two Phase-Locked Loop Chips

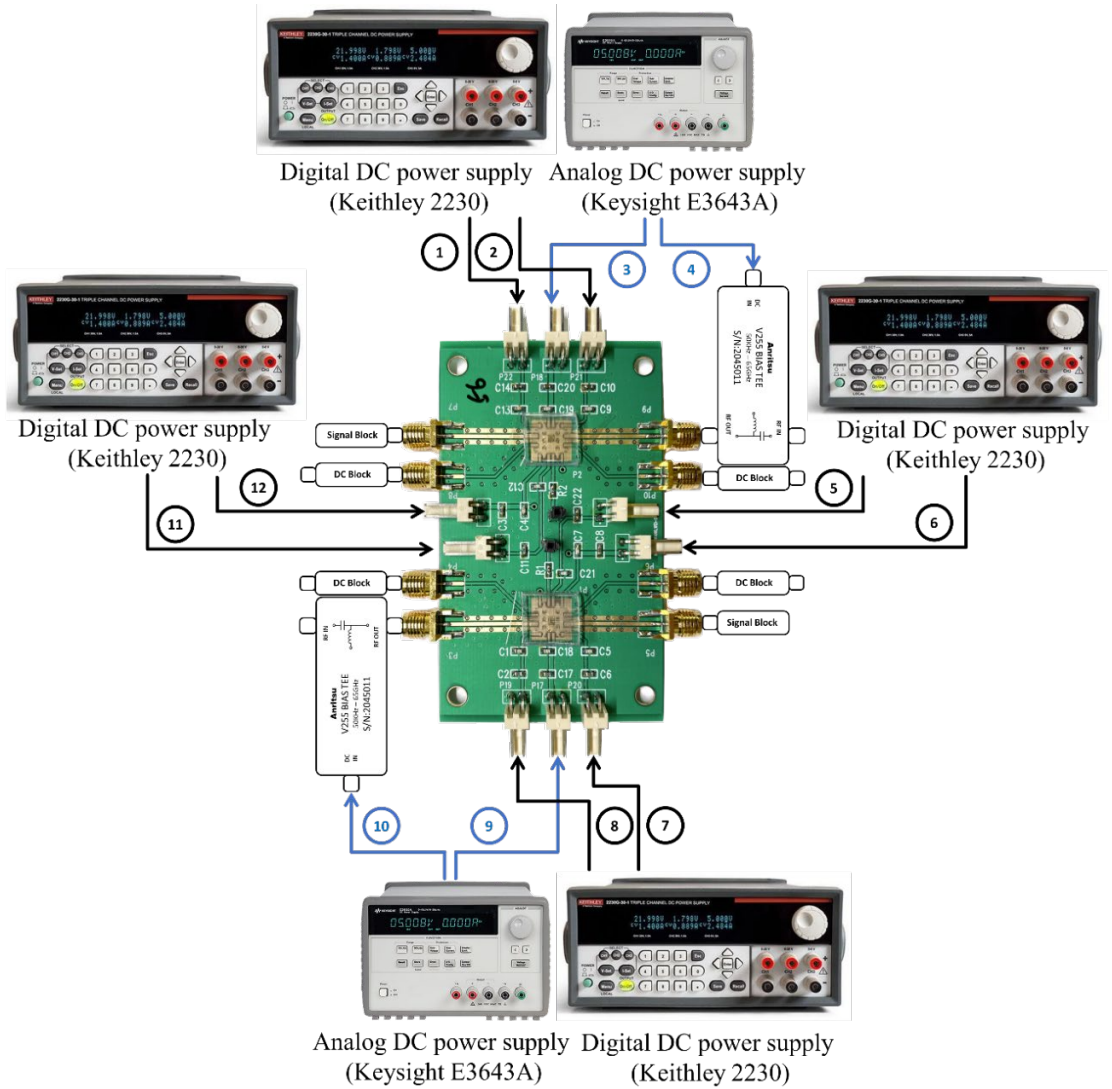


Fig 47. PCB Board power terminal of measurement

Tab 3. PCB Board analog power terminal of measurement

Number	3	4	9	10
Analog (V)	1.8	0.9	1.8	0.9

Tab 4. PCB Board digital power terminal of measurement

Number	1	2	5	6	7	8	11	12
Digital (V)	1.2	0.6	0-1.5	1.8	1.2	0.6	0-1.5	1.8

PCB Board with Two Phase-Locked Loop Chips

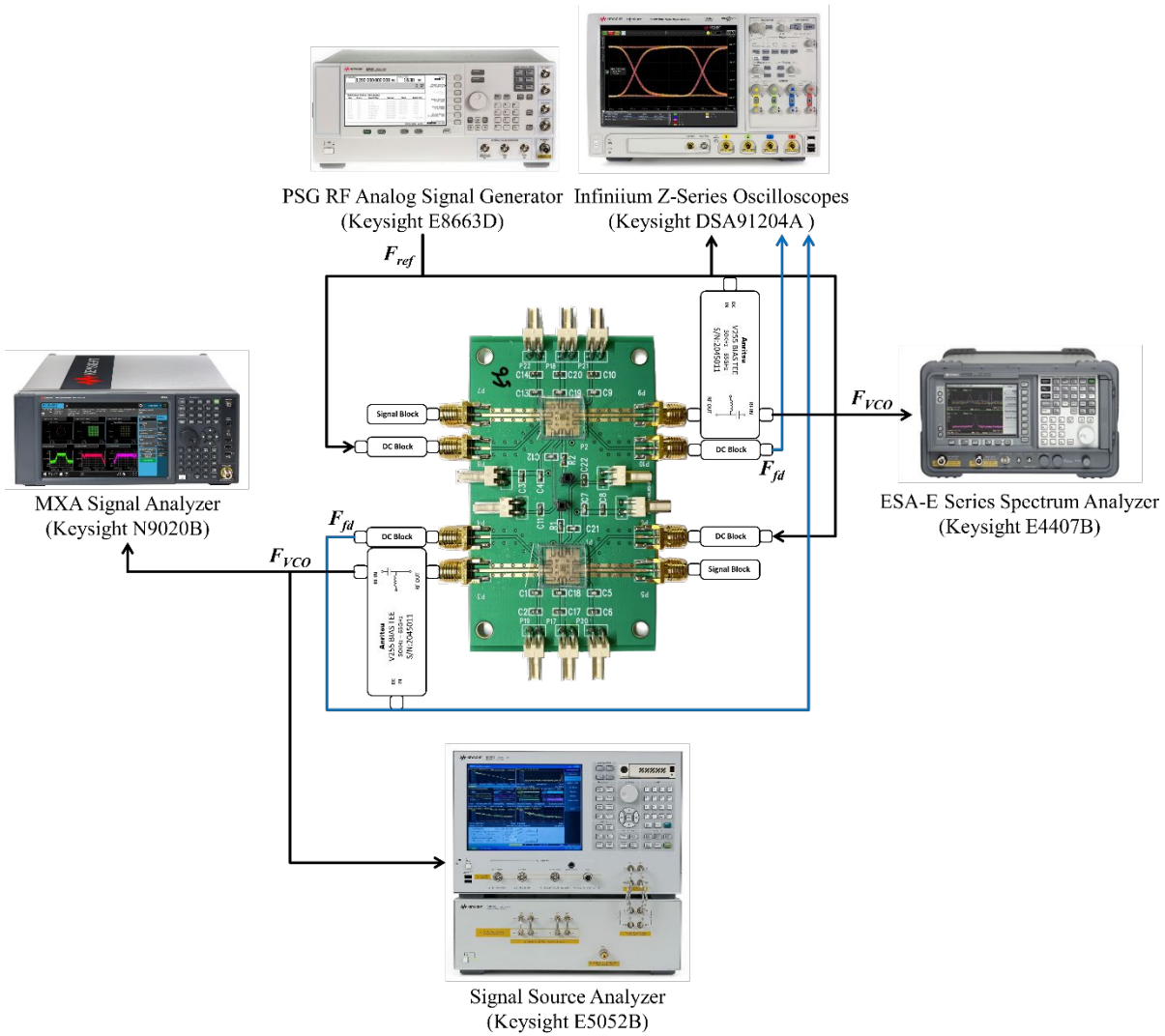


Fig 48. PCB Board signal terminal of measurement

Measurement

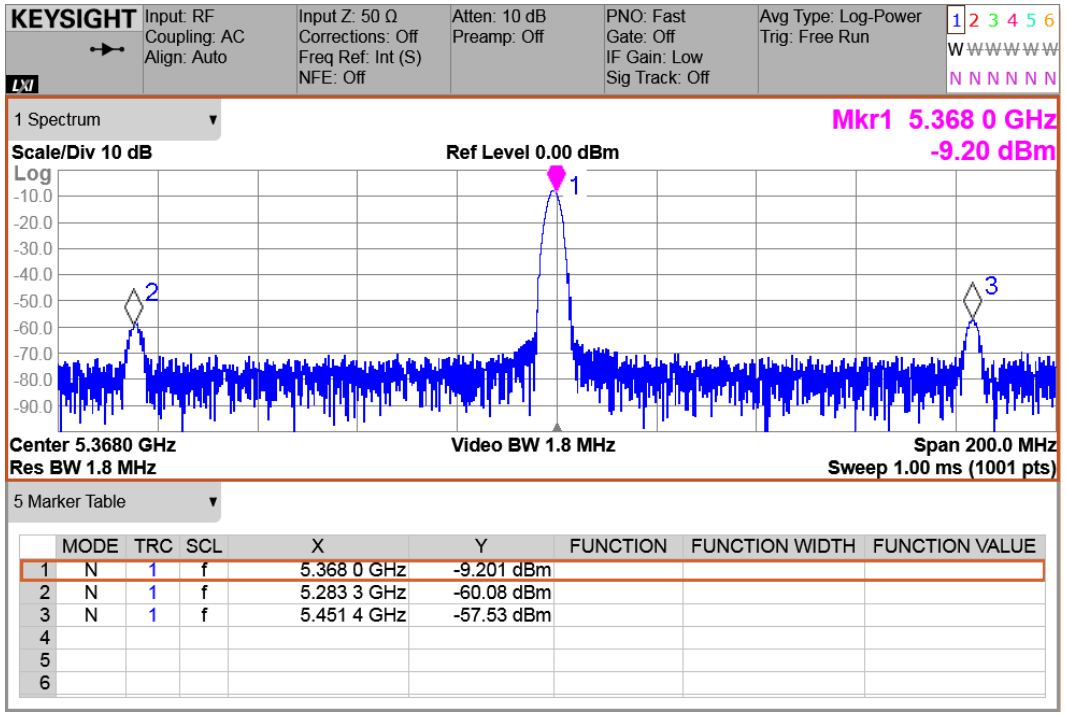


Fig 49. VCO1 circuit output power

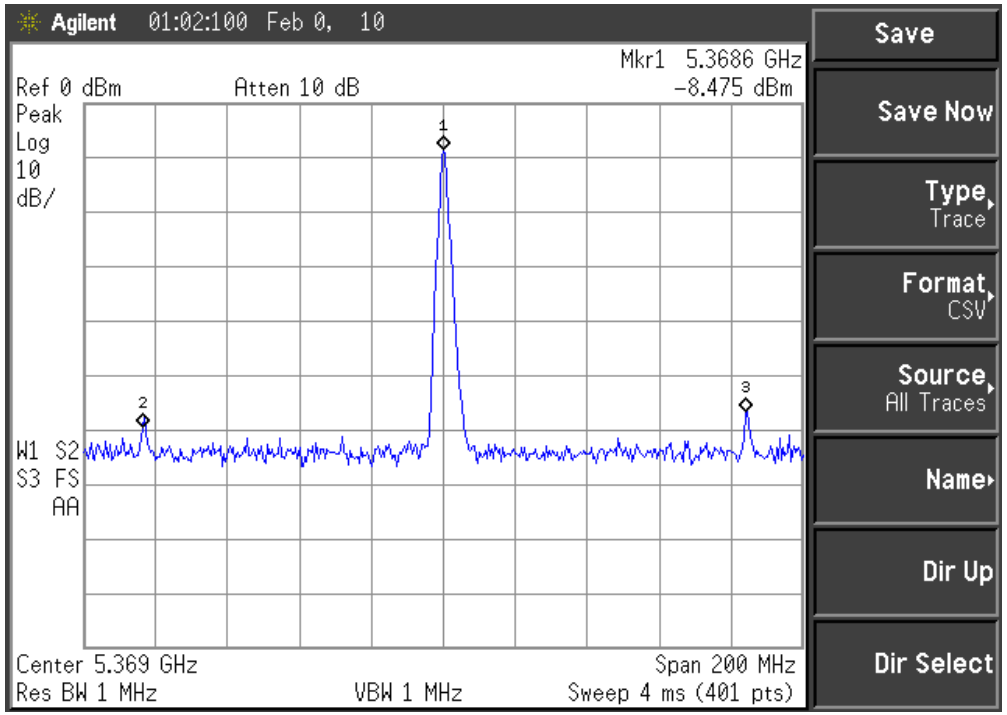


Fig 50. VCO2 circuit output power

Measurement

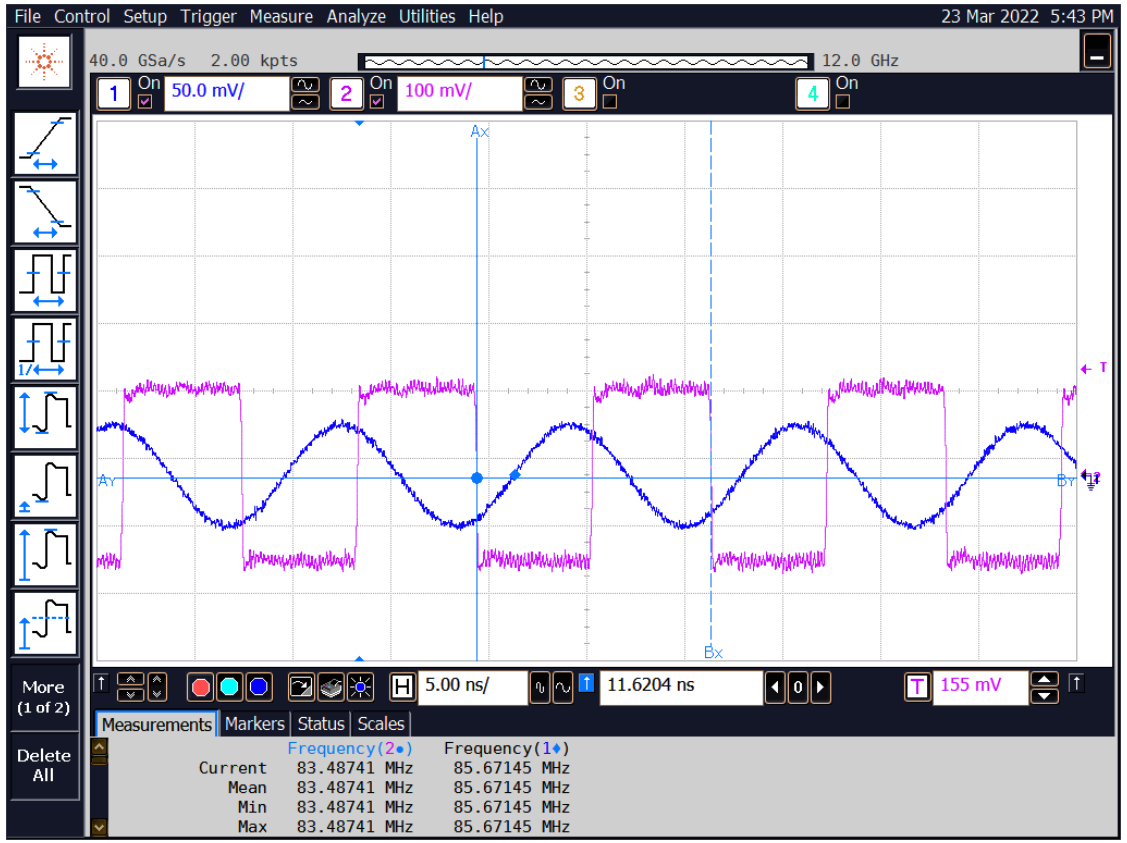


Fig 51. F_{ref} and F_{fd} of VCO circuit

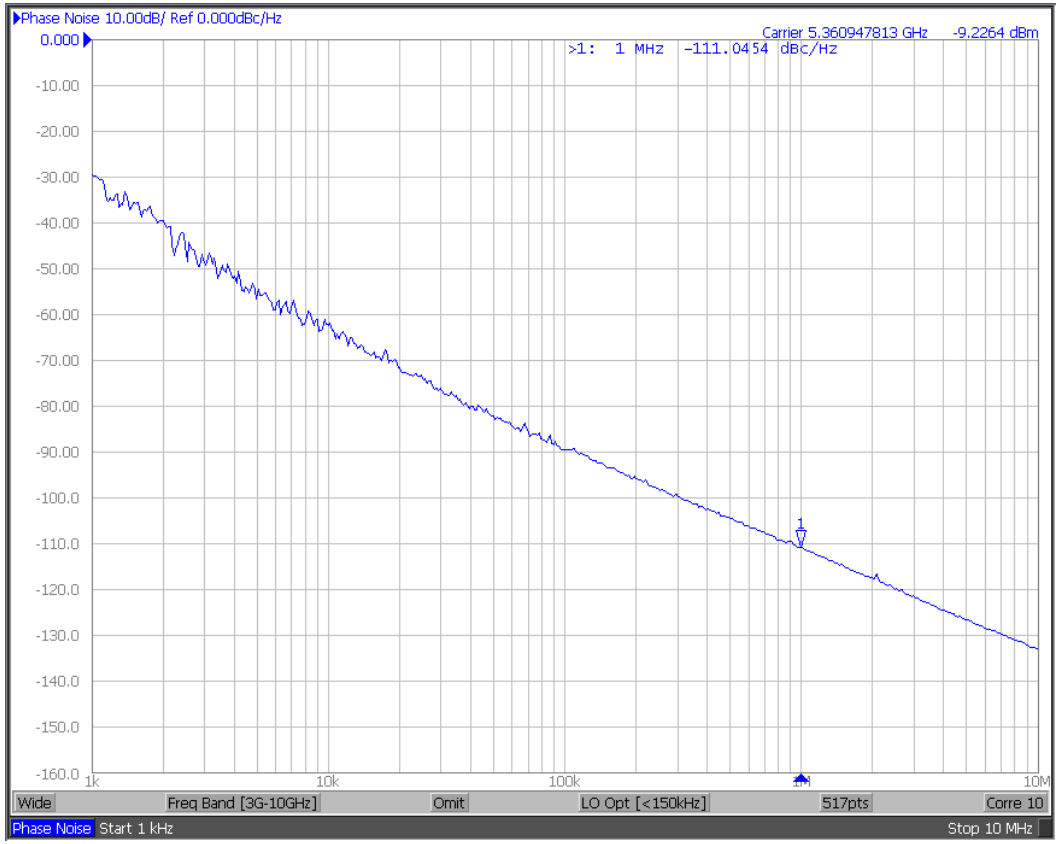


Fig 52. Phase noise of VCO circuit

Comparison Table

Tab 5. Circuit performance comparison Table

	This work	[35]	[36]	[37]
CMOS Tech. (nm)	180	180	180	130
Supply Voltage	1.8 V	1.8 V	1.8 V	1.2 V
Current Dissipation	9.5 mA	11 mA	9.5 mA	22.7 mA
Division	64	8/10	4	Frac-N
Reference Frequency (MHz)	83.75-92.96	10	4.7	50
Operation Frequency (GHz)	5.36-5.95	5.27-5.6	4.9-5.3	2.3-2.4
Locking Time (ms)	< 6	< 20	<10	
Controllable Delay Time (ps)	63-203	-	-	-
Controlled-voltage of VCDL (V)	0~1.8	-	-	-
Phase Noise @1MHz (dBc/Hz)	-111	-114	-118	-120
Reference Spurs (dBc)	<-48	<-70	<-71	<-72
Chip Area (mm^2)	0.984	1.610	1.61	-

Layout of Quadrature Voltage Controlled Oscillator

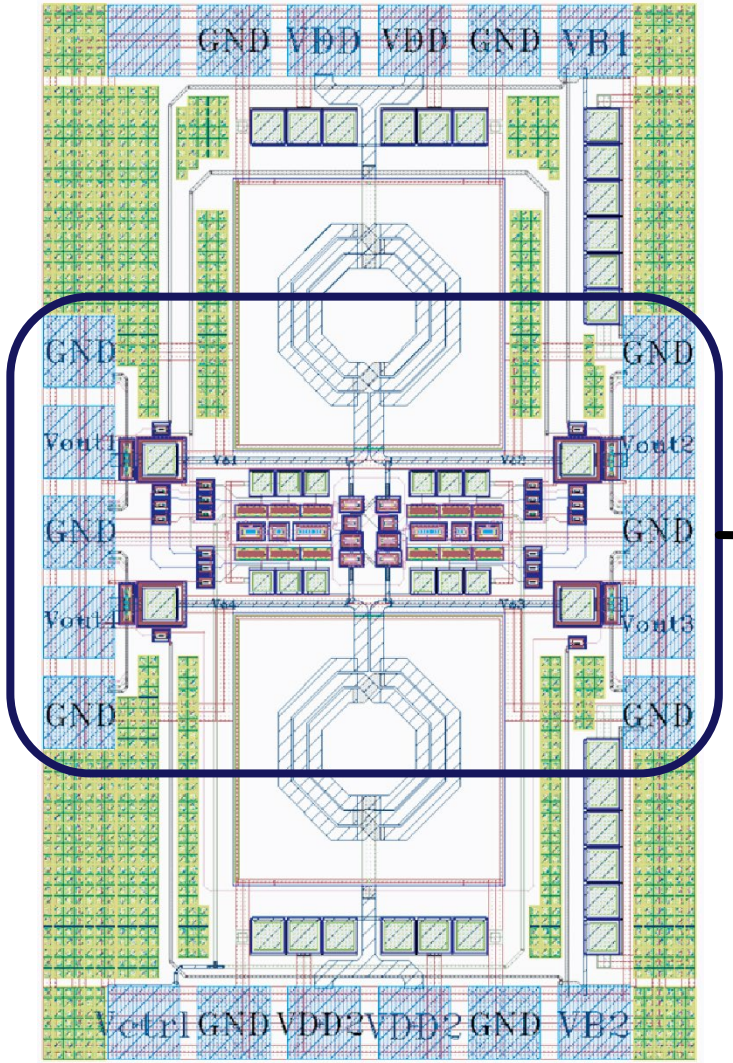


Fig 53. Layout of QVCO

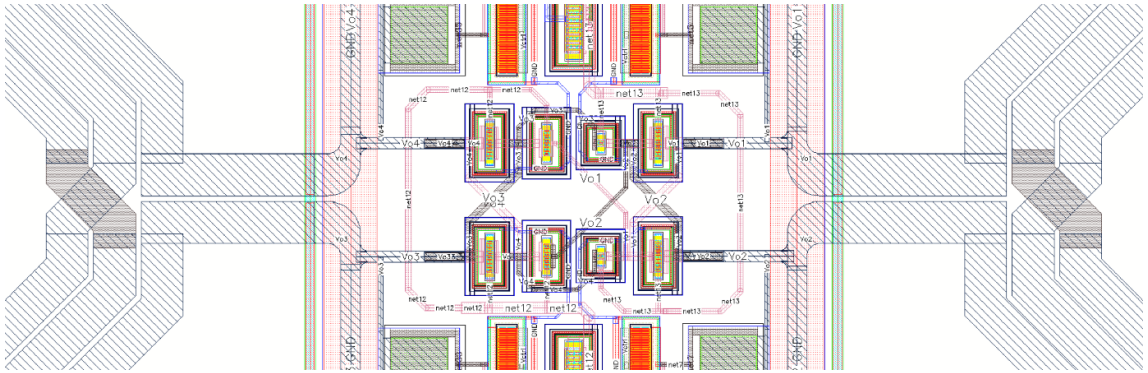


Fig 54. QVCO circuit in detail view

Post Simulation of Quadrature Voltage Controlled Oscillator

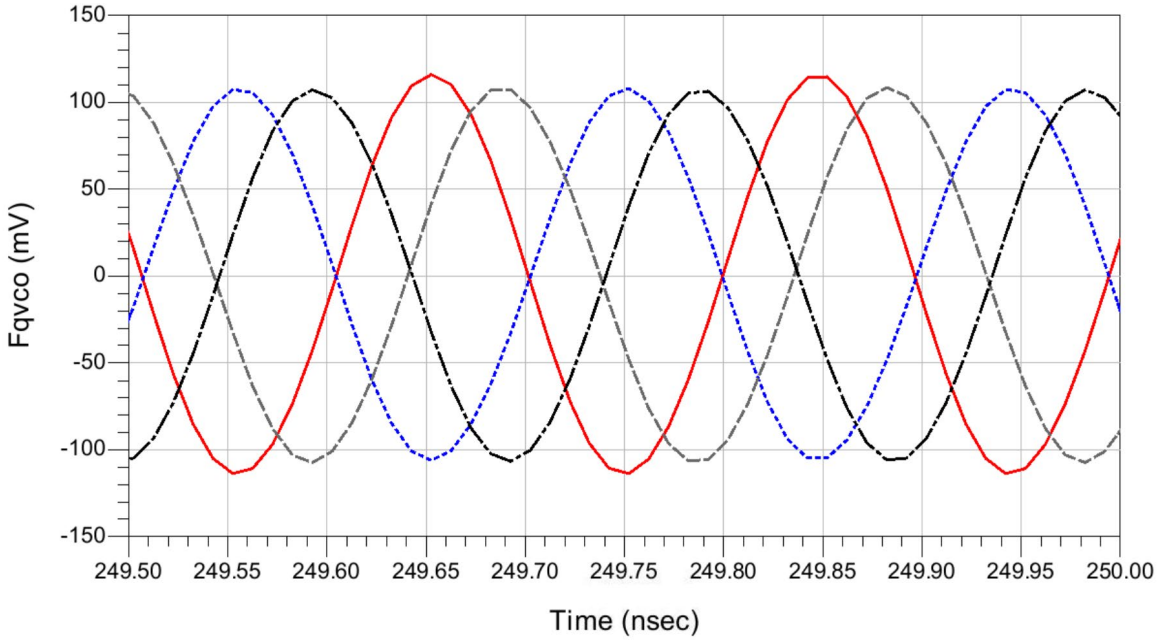


Fig 55. QVCO circuit output

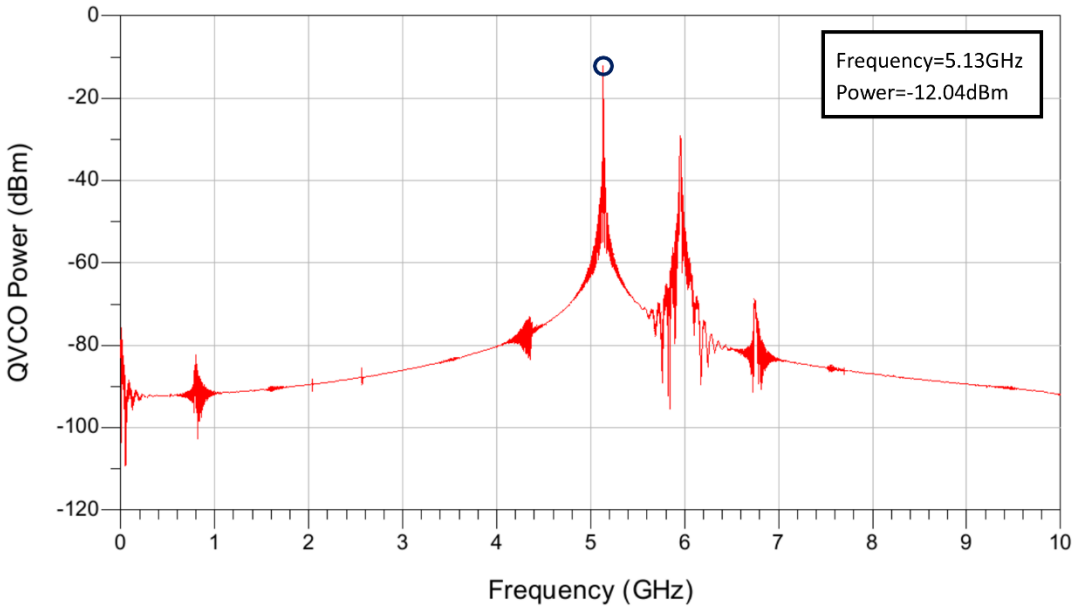


Fig 56. QVCO circuit output power

Post Simulation of Quadrature Voltage Controlled Oscillator

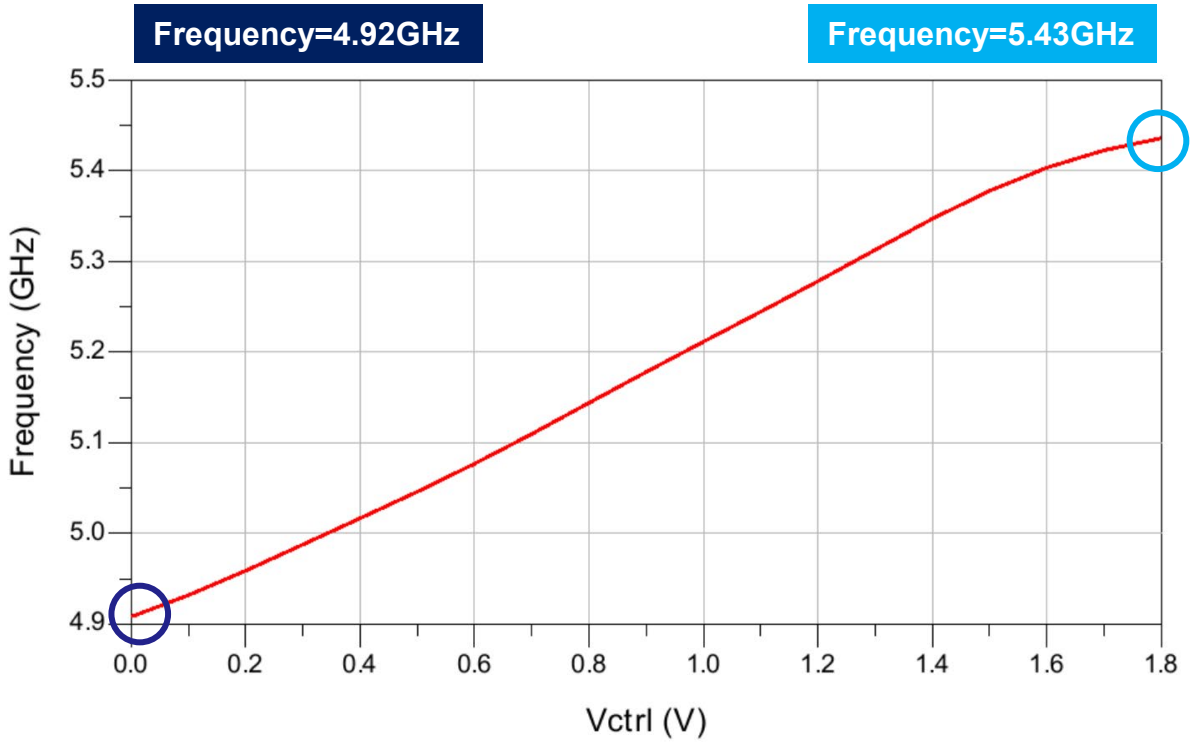


Fig 57. Tuning range of QVCO circuit

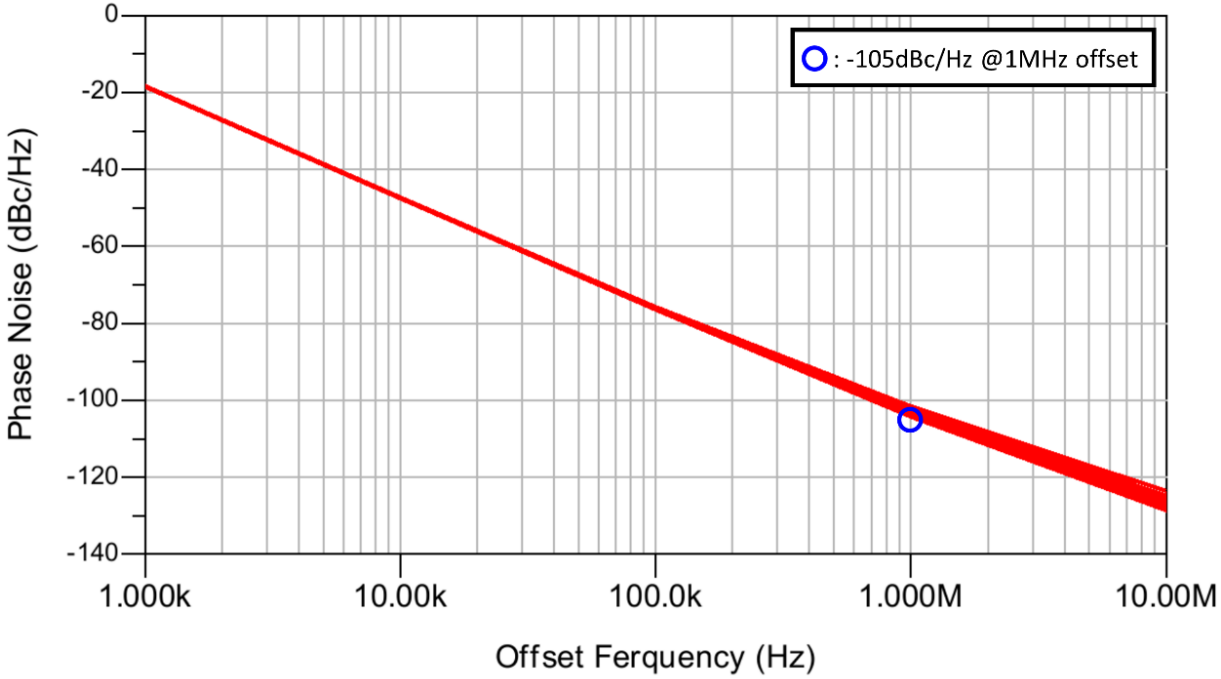


Fig 58. QVCO circuit phase noise

I/O Pin of Quadrature Voltage Controlled Oscillator

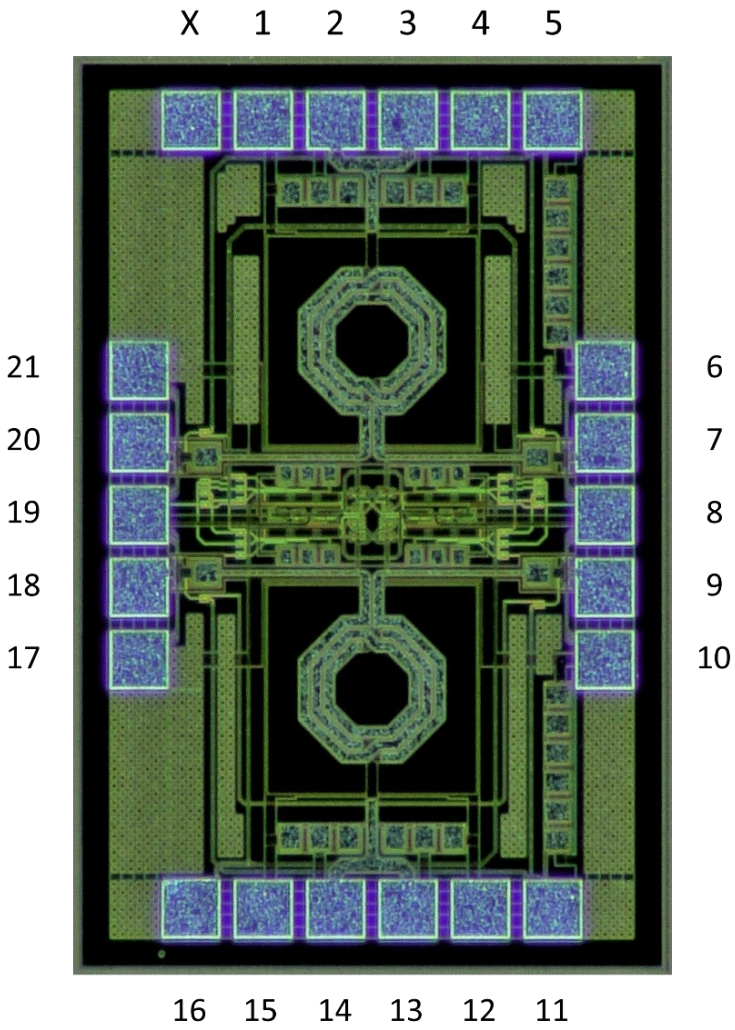


Fig 59. Photomicrograph of QVCO

Tab 6. 3GPP 5G NR FR1 add to Release 16

1	*GND*	Ground for chip	12	*GND*	Ground for chip
2	*VDD*	Supply for QVCO Circuit	13	*VDD2*	Supply for QVCO Circuit
3	*VDD*	Supply for QVCO Circuit	14	*VDD2*	Supply for QVCO Circuit
4	*GND*	Ground for chip	15	*GND*	Ground for chip
5	*VB1*	Bias for buffer	16	*Vctrl*	Test for Vctrl
6	*GND*	Ground for chip	17	*GND*	Ground for chip
7	*Vout2*	Output for QVCO	18	*Vout4*	Output for QVCO
8	*GND*	Ground for chip	19	*GND*	Ground for chip
9	*Vout3*	Output for QVCO	20	*Vout1*	Output for QVCO
10	*GND*	Ground for chip	21	*GND*	Ground for chip
11	*VB2*	Bias for buffer	X	*NO*	Unused

On Wafer Measurement of QVCO

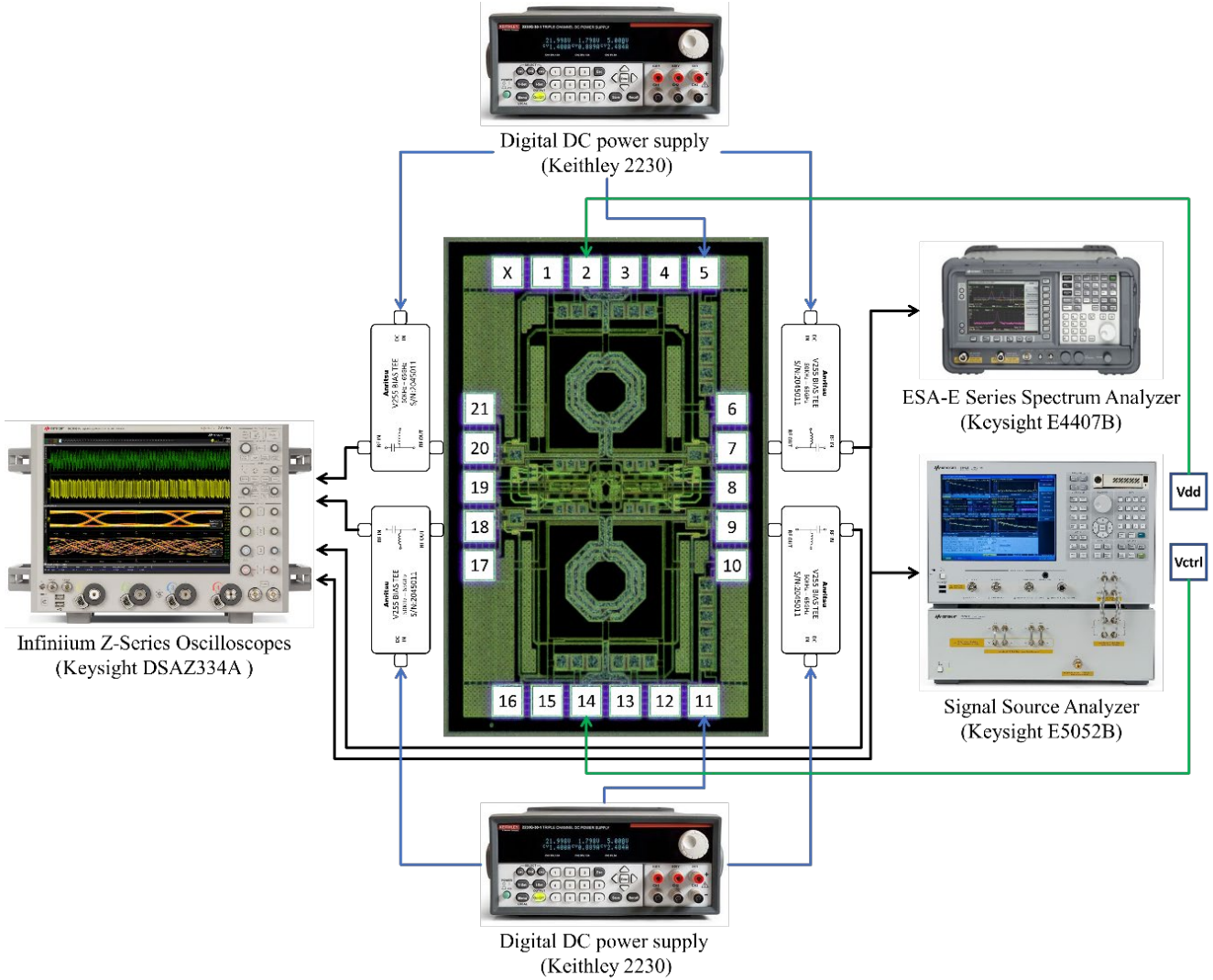


Fig 60. QVCO input power and output signal terminal of measurement

On Wafer Measurement of QVCO

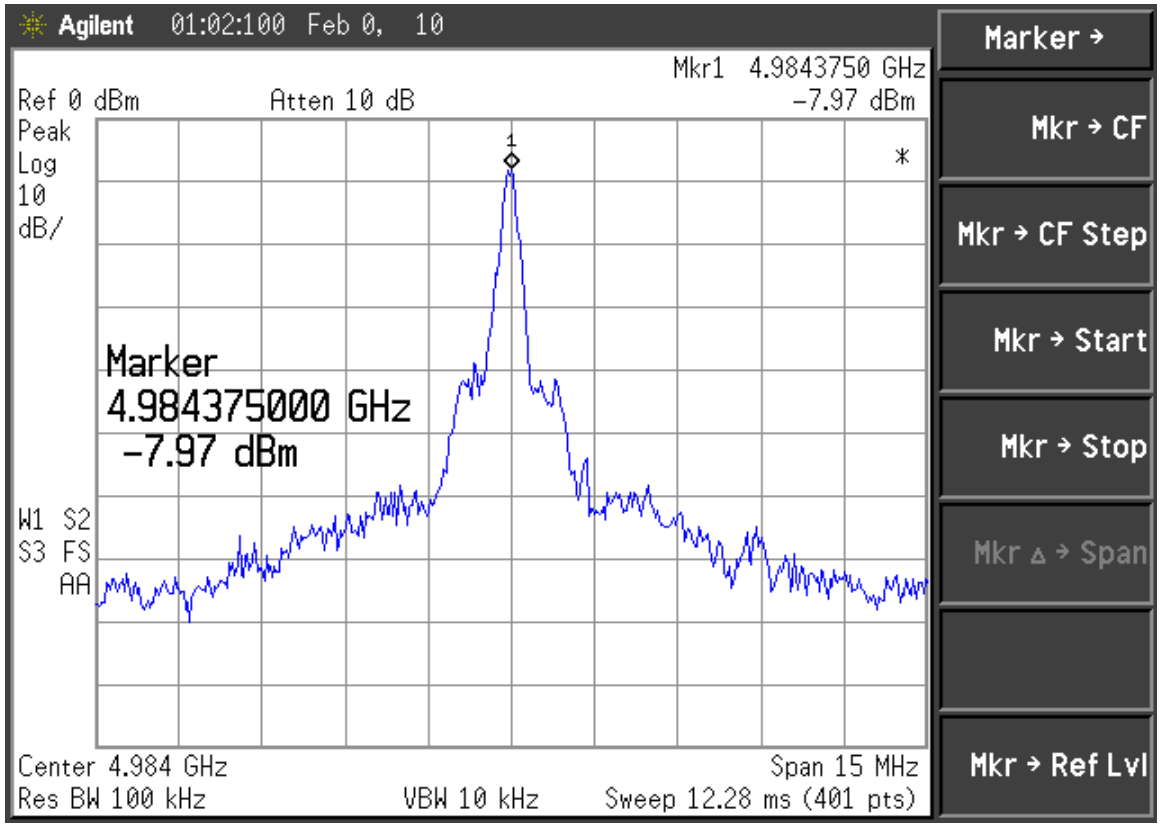


Fig 61. QVCO circuit output power

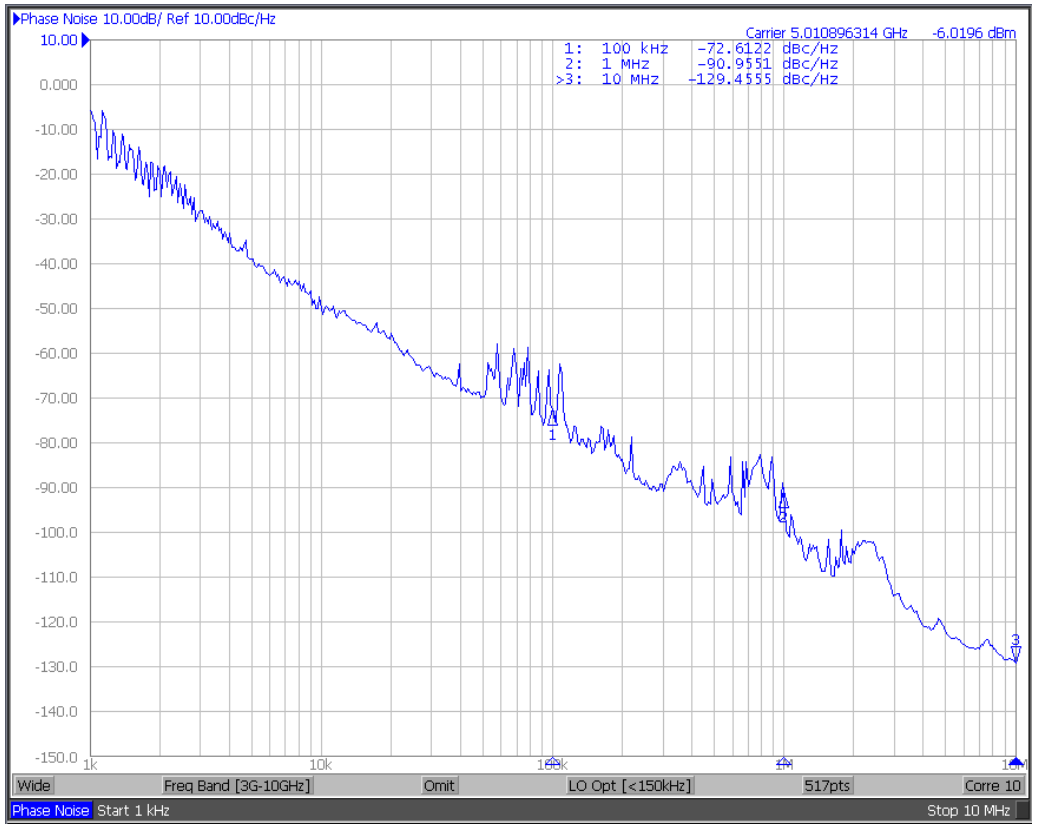


Fig 62. QVCO phase noise

On Wafer Measurement of QVCO

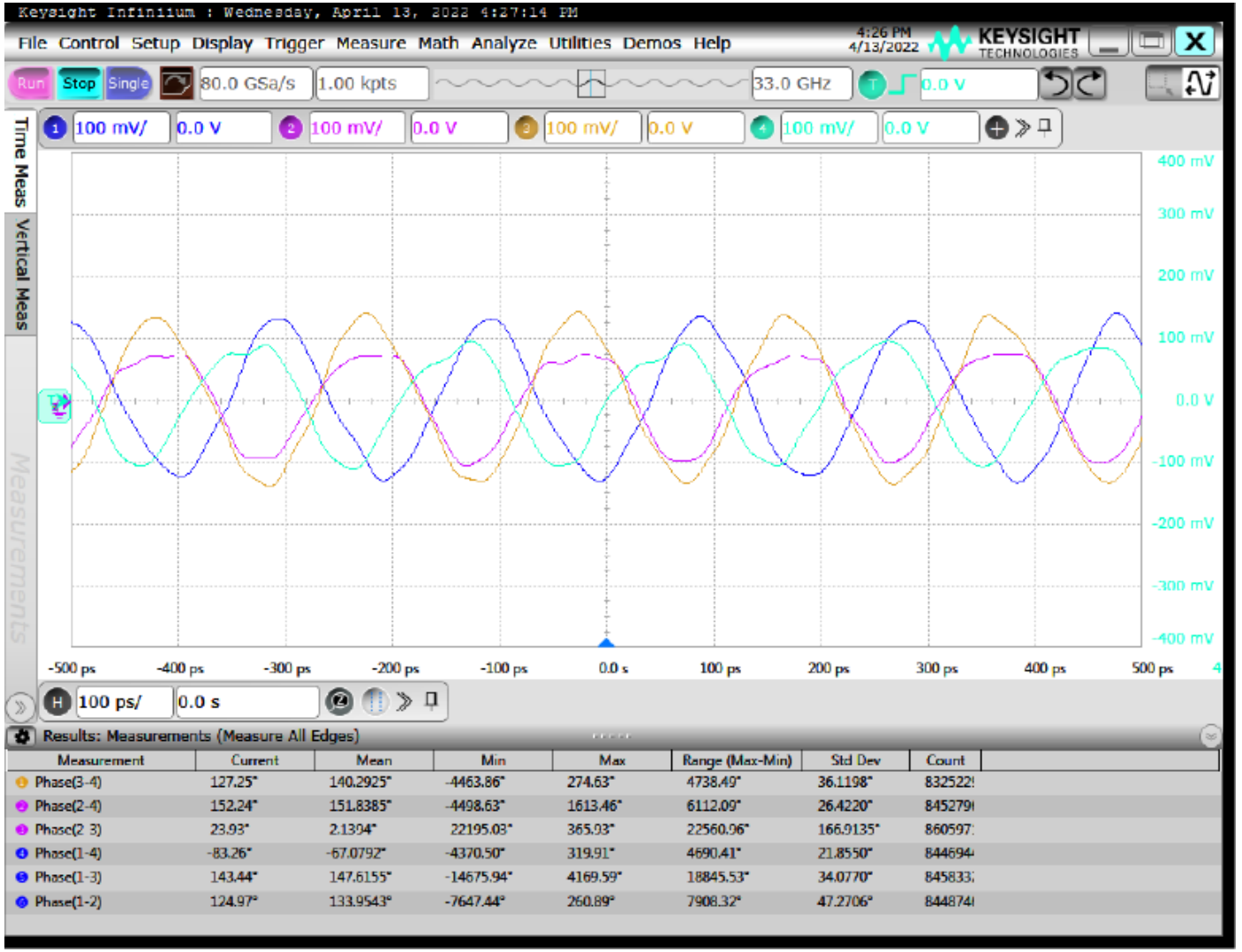


Fig 63. QVCO circuit output signal

Contents

- Introduction
- Related Technology
- Phase Frequency Detector & Charge pump
- Loop Filter
- Voltage Controlled Oscillator
- Quadrature-Voltage Controlled Oscillator
- Frequency Divider
- Varactor Delay Element
- Layout & Measurement
- **Conclusion & Reference**

Conclusion

- Promote the conformability of the chip
  - Put the loop filter in the chip promote the device integration.
  - Supply other control variable V_{ctrl2} to the communication system.

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證明文件掃描



單晶片認證



TQC認證



專題製作



論文製作



競賽獎狀



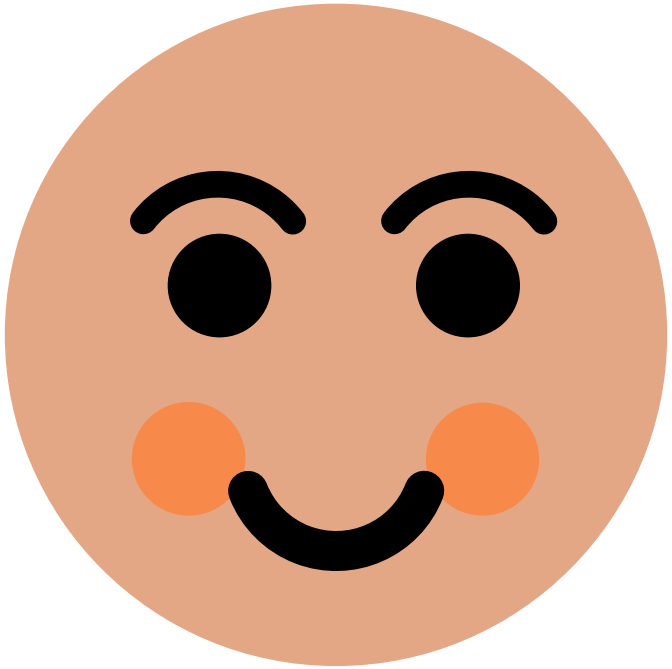
多益成績



新尖兵-陽明交大



TSRI課程



Thank you for your attention ! ! !